

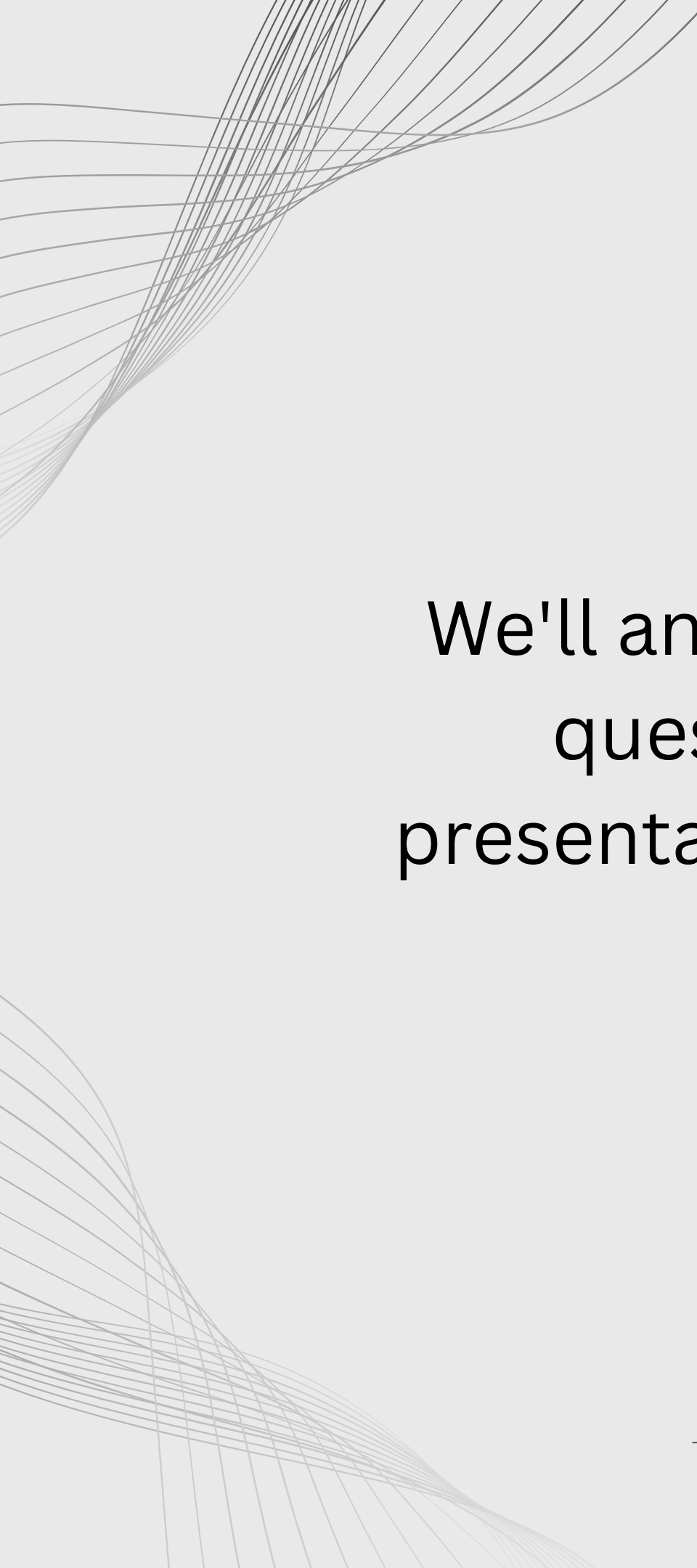
TENNIS GAMES ANALYSIS

By

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Mozhgan Mohammadi



We'll analyze tennis games dataset by exploring 17 key questions from a large dataset of matches. Our presentation will highlight the insights we've uncovered from this rich information.

QUESTION 1 :

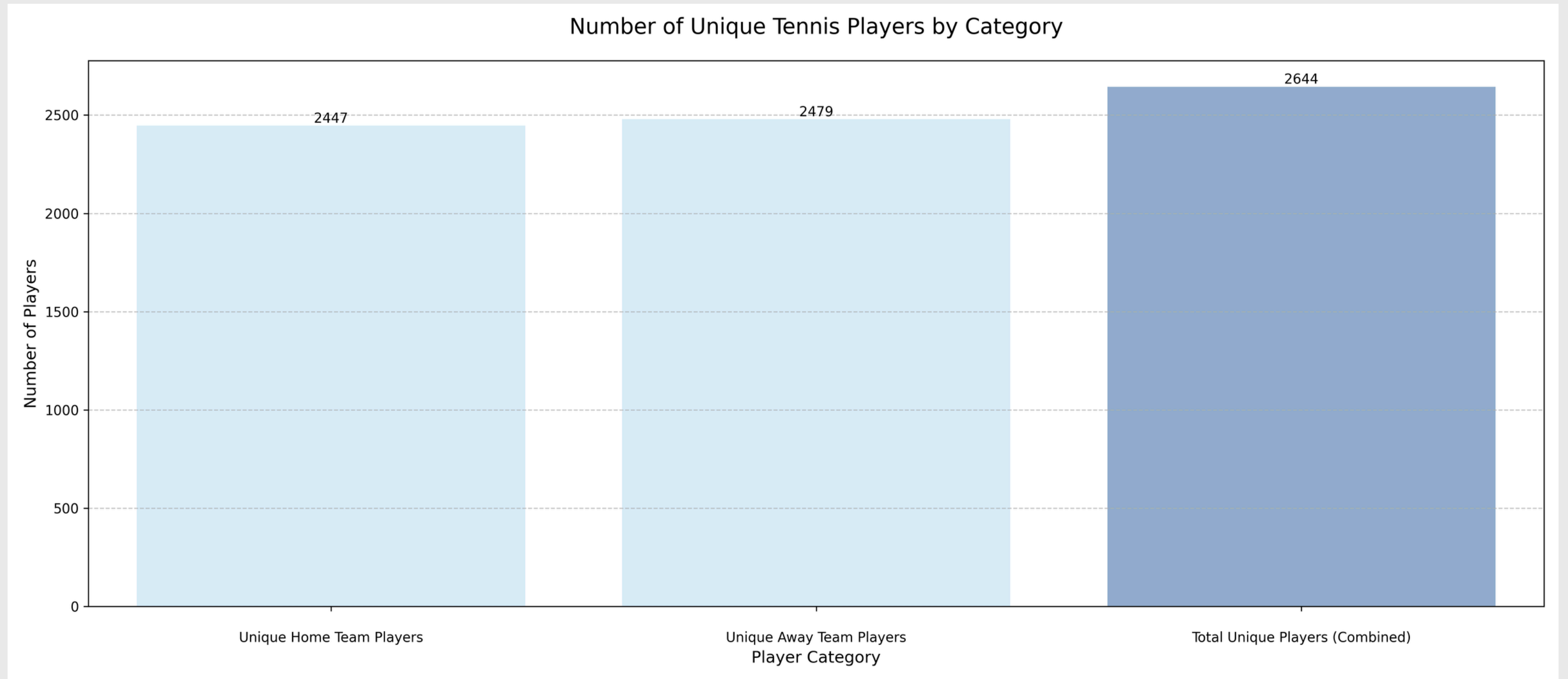
How many tennis players are included in the dataset?

Number of Tennis Players in the Dataset (Excluding Nulls):

Unique Home Team Players: **2447**

Unique Away Team Players: **2479**

Total Unique Players (Combined): **2644**



QUESTION 2 :

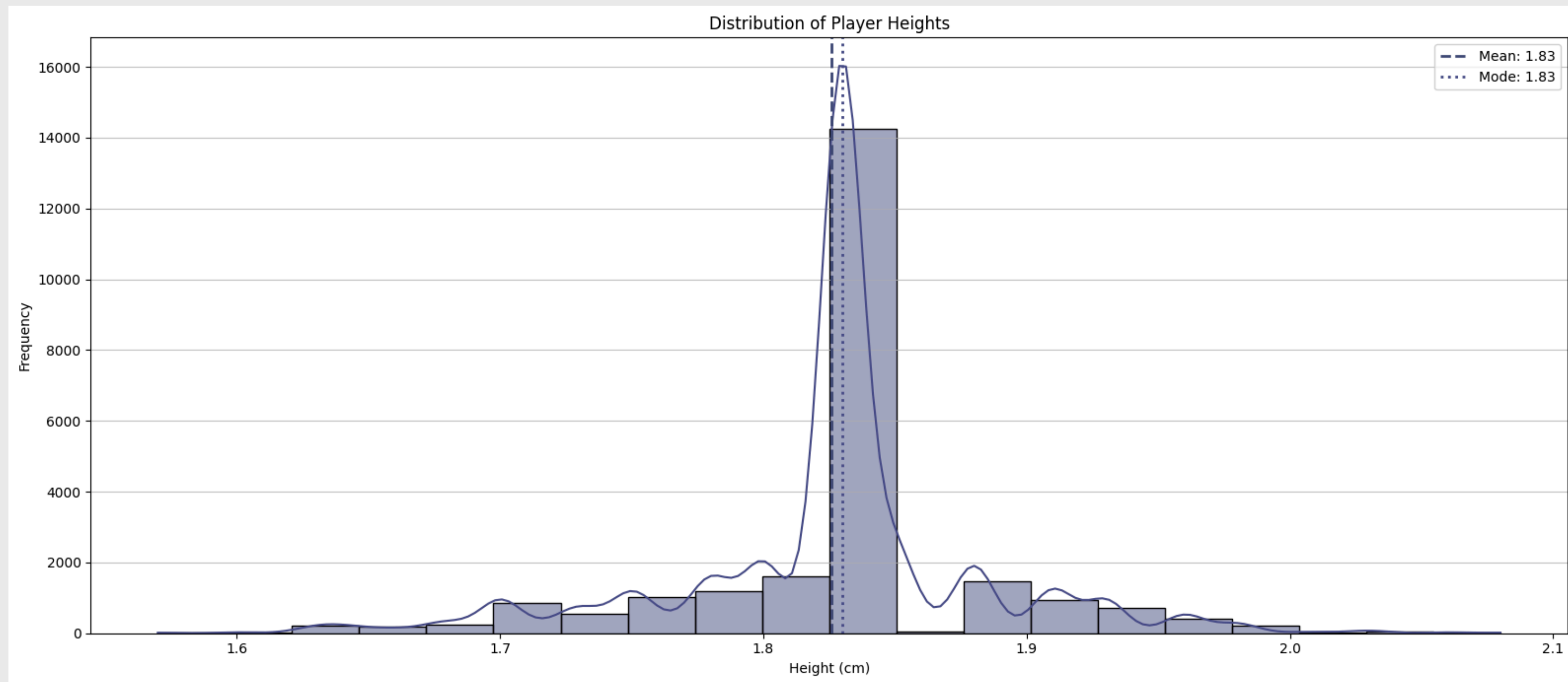
What is the average height of the players?

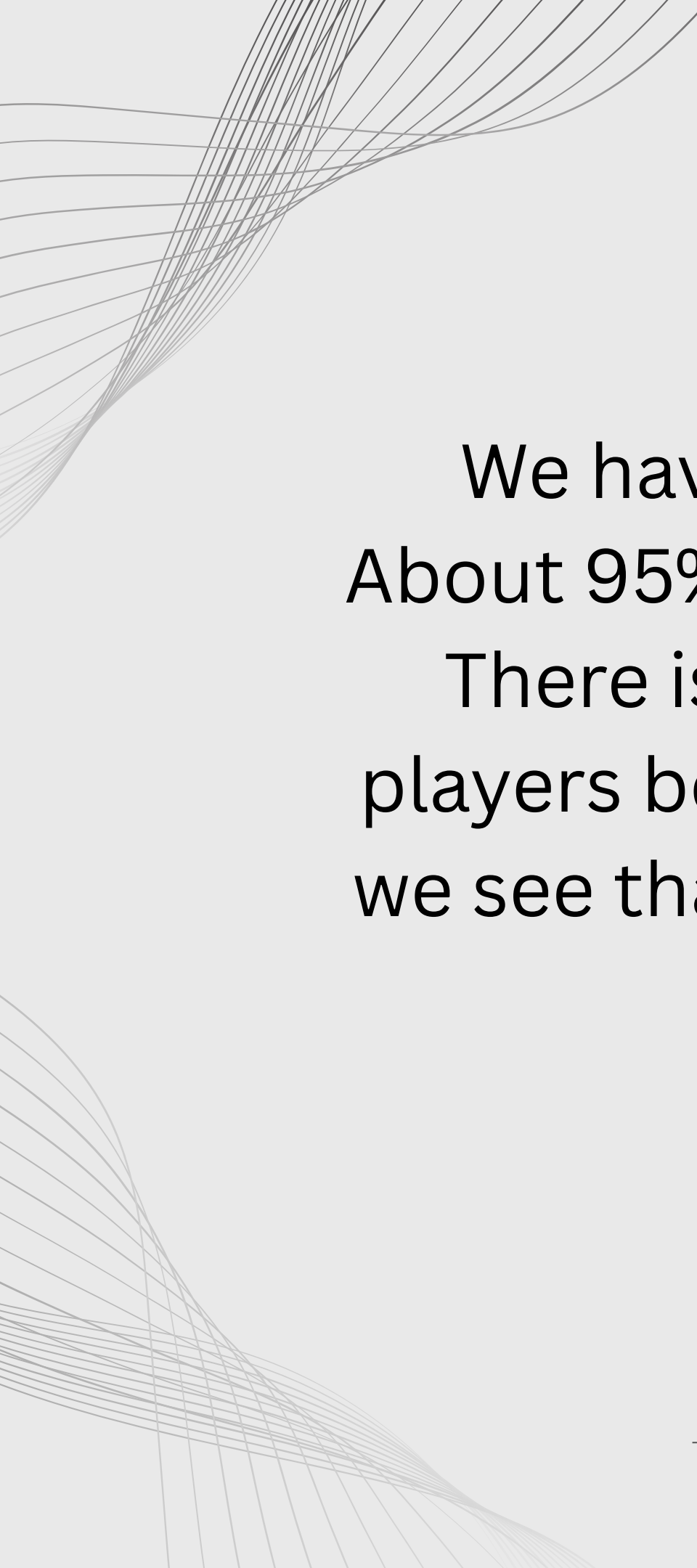
Combined all players data shape:

Median height used for filling NaN values: **1.83m**

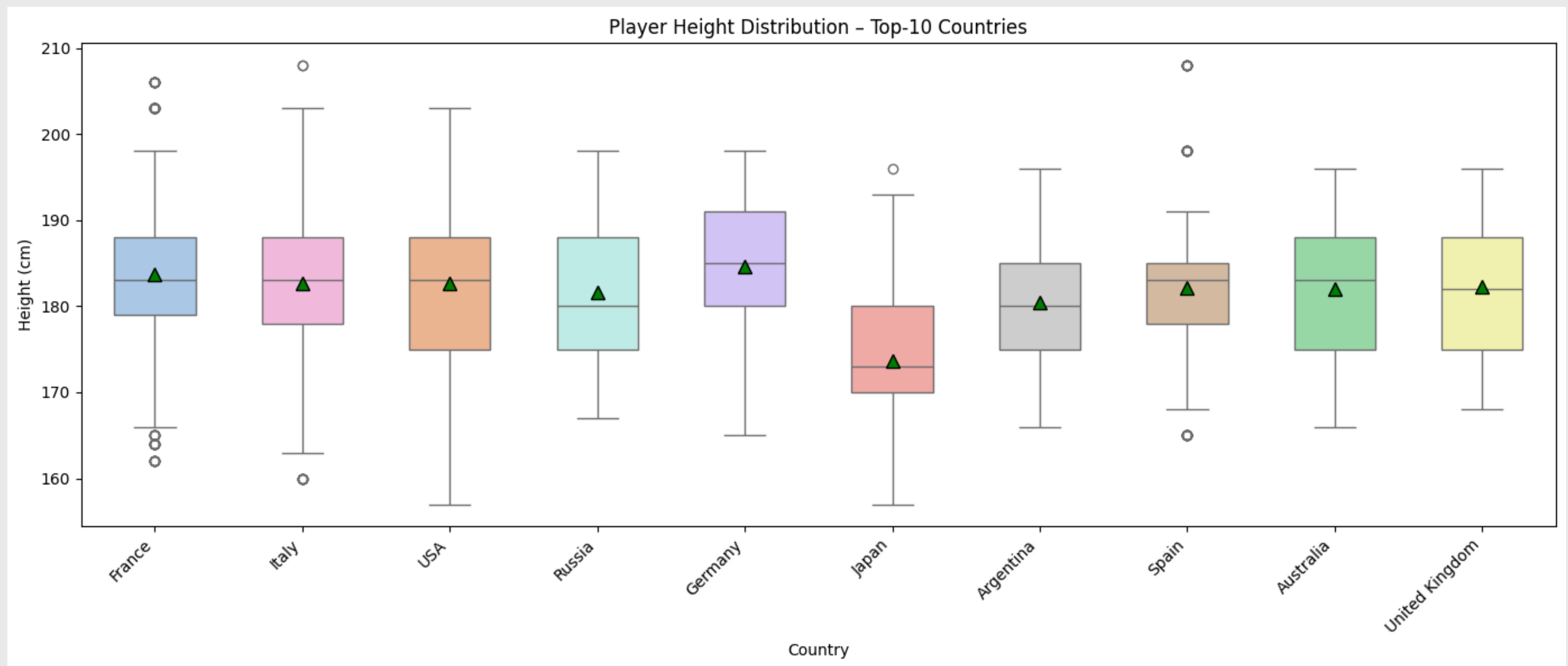
The Arithmetic Mean of Player Heights is: **1.83m**

Mode of Height: **1.83m**





We have a tall, narrow bell curve peaking at **1.83 cm**.
About 95% of all players fall between **1.78 cm** and **1.88 cm**.
There is a light left skew, which shows there are more players below 1.83 cm than above it and if we look closely we see that the tail stretches slightly more to the left than to the right.



QUESTION 3 :

Which player has the highest number of wins?

Total Actual Winners Found: **2320**

Top Winning Players:

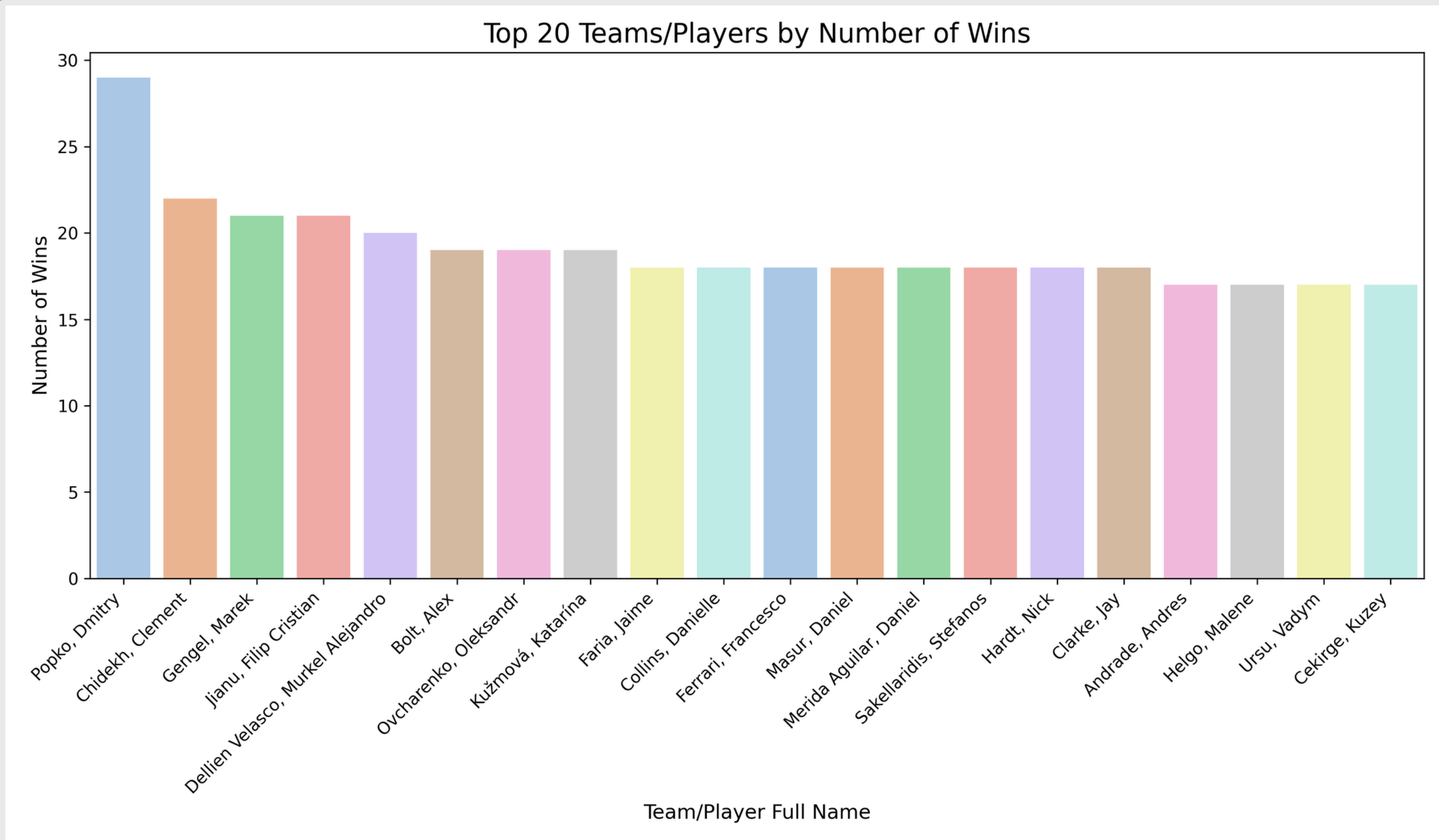
Popko, Dmitry **29**

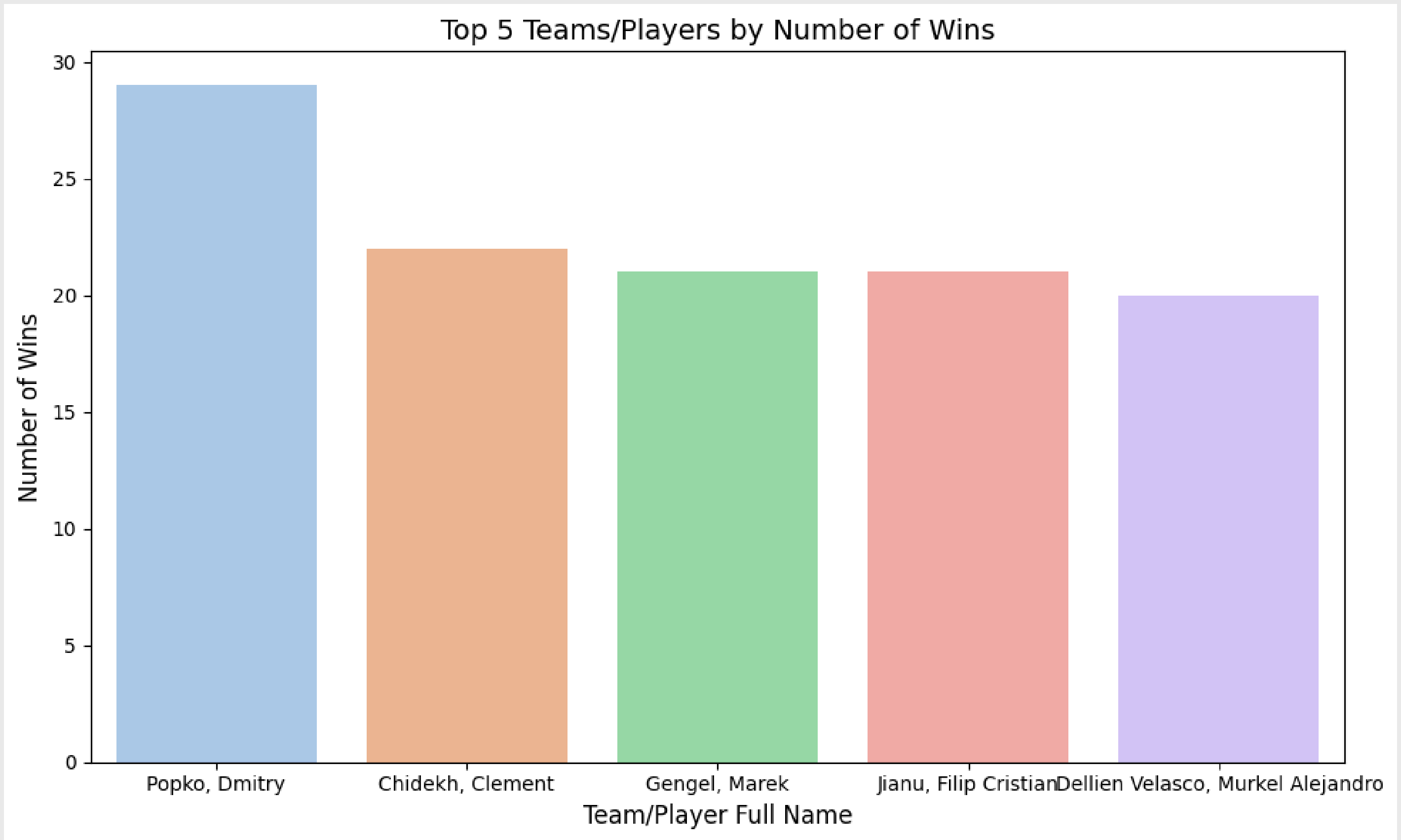
Chidekh, Clement **22**

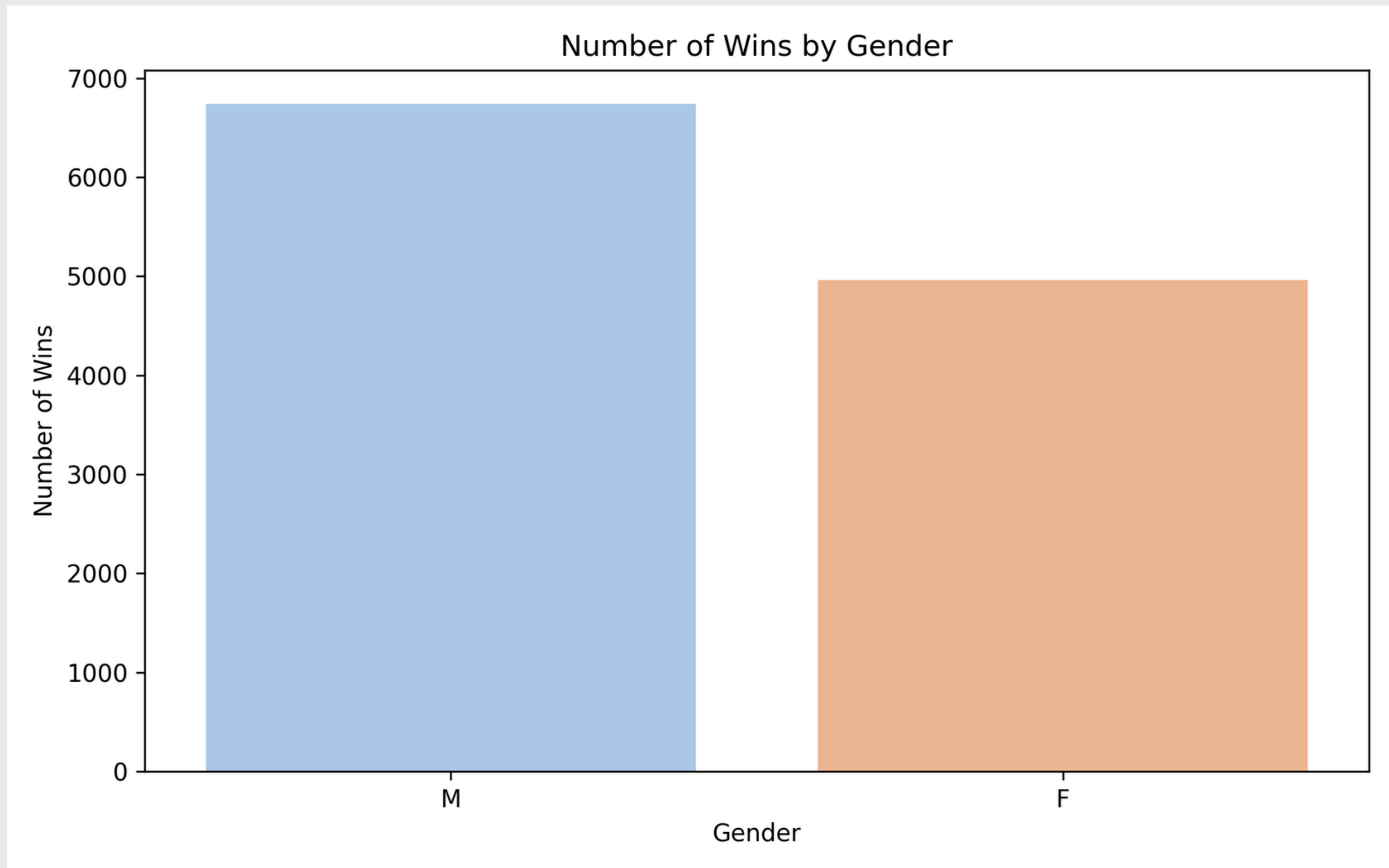
Gengel, Marek **21**

Jianu, Filip Cristian **21**

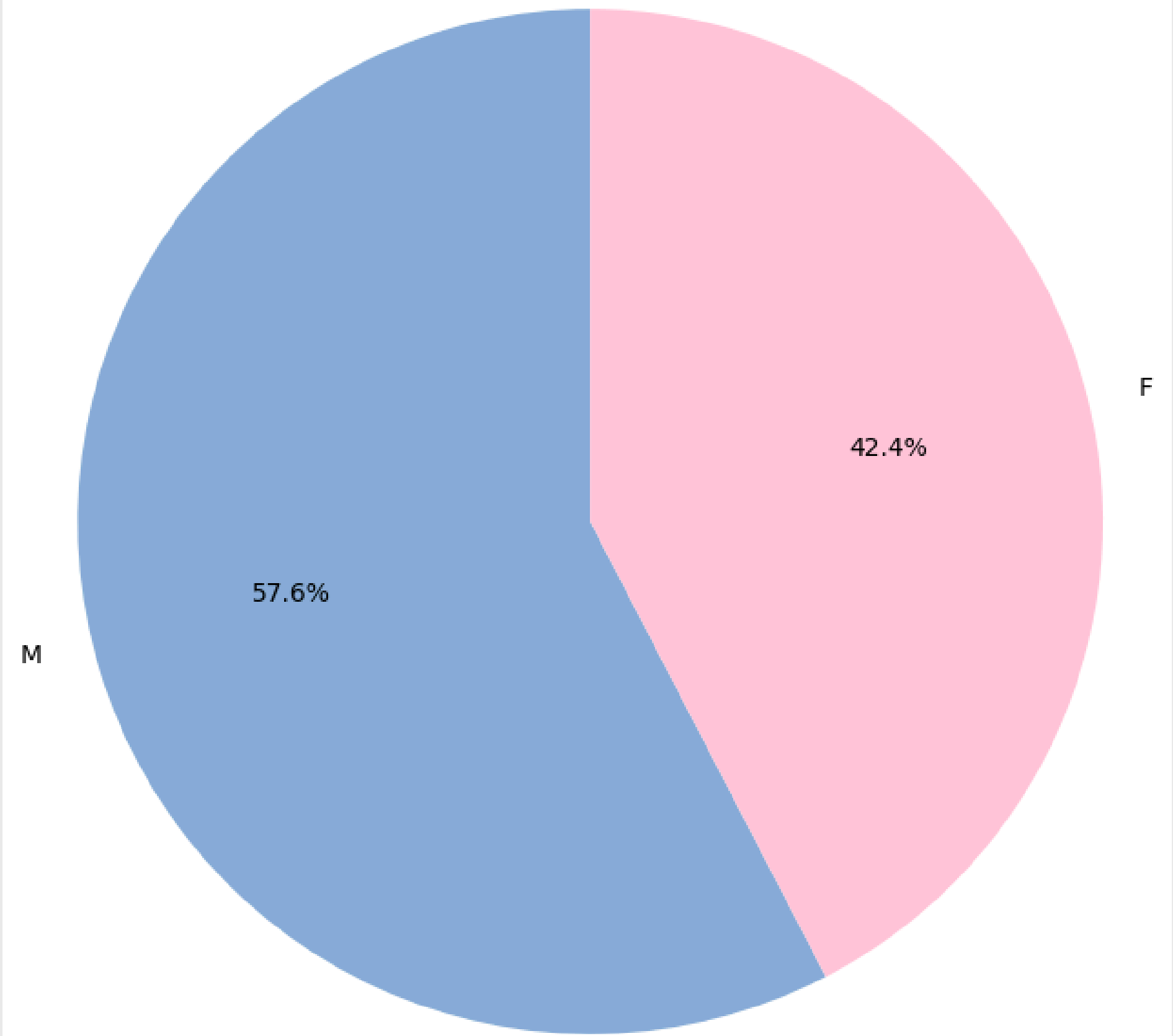
Dellien Velsco, Murkel Alejandro **20**







Percentage of Wins by Gender

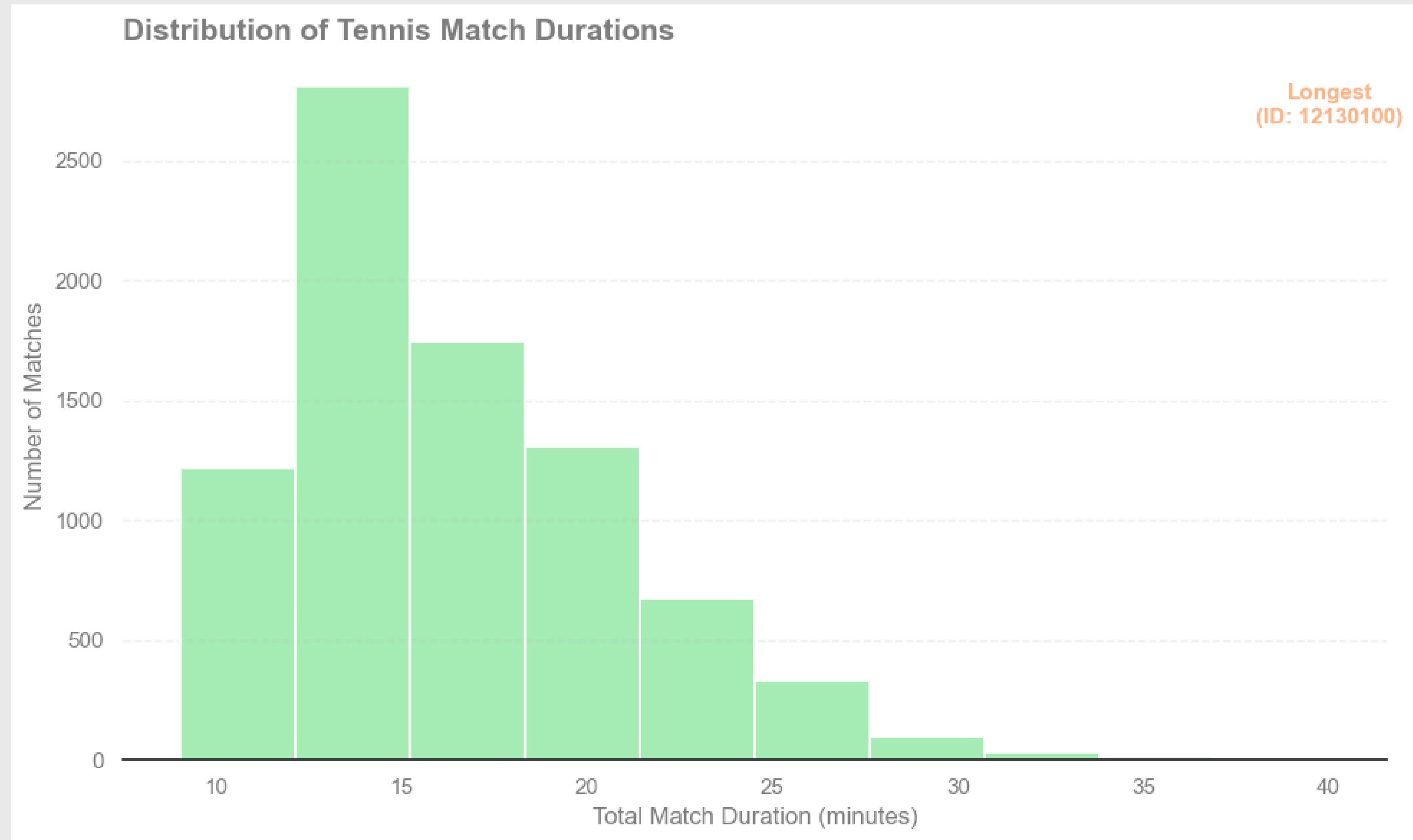


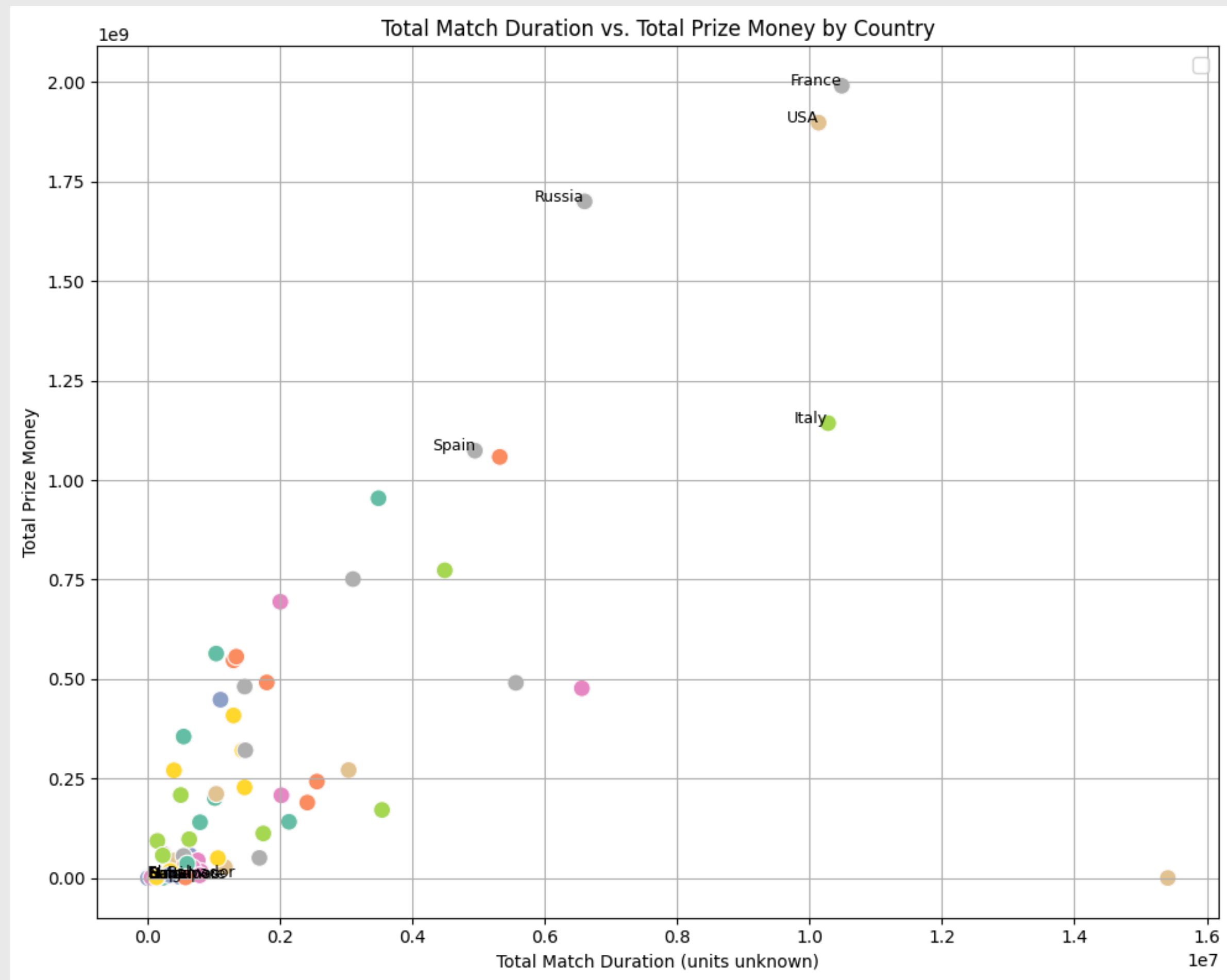
QUESTION 4 :

What is the longest match recorded in terms of duration?

ID of the Longest Match: **12130100**

Duration: **02:52:01**



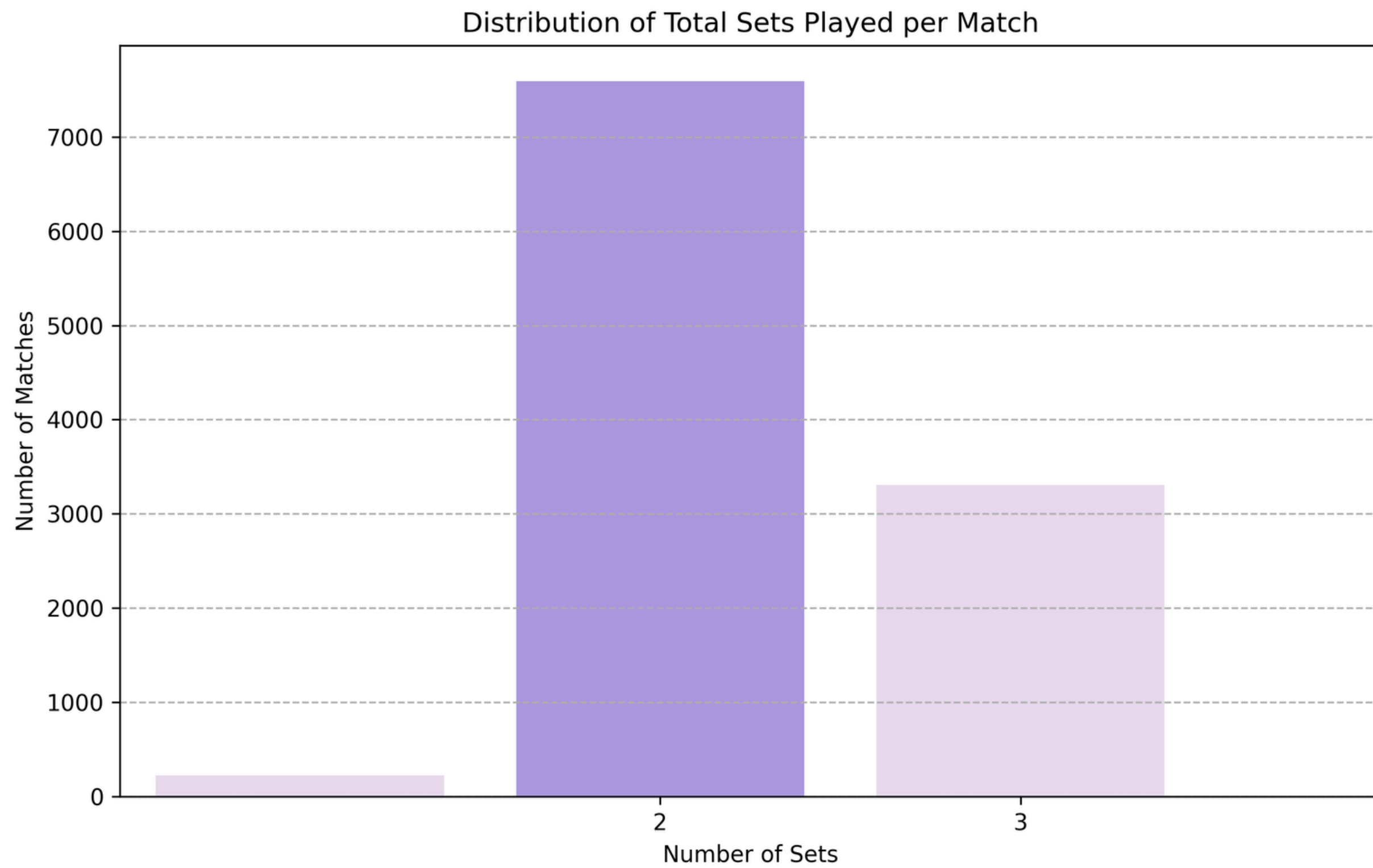


QUESTION 5 :

How many sets are typically played in a tennis match?

The average number of sets played per match is: **2.28**

and most of matches are **2** sets



QUESTION 6 :

Which country has produced the most successful tennis players?

1 - France 992

2 - Italy 930

3 - USA 889

Top 10 countries by number of match wins:

France **992**

Italy **930**

USA **889**

Russia **621**

Germany **557**

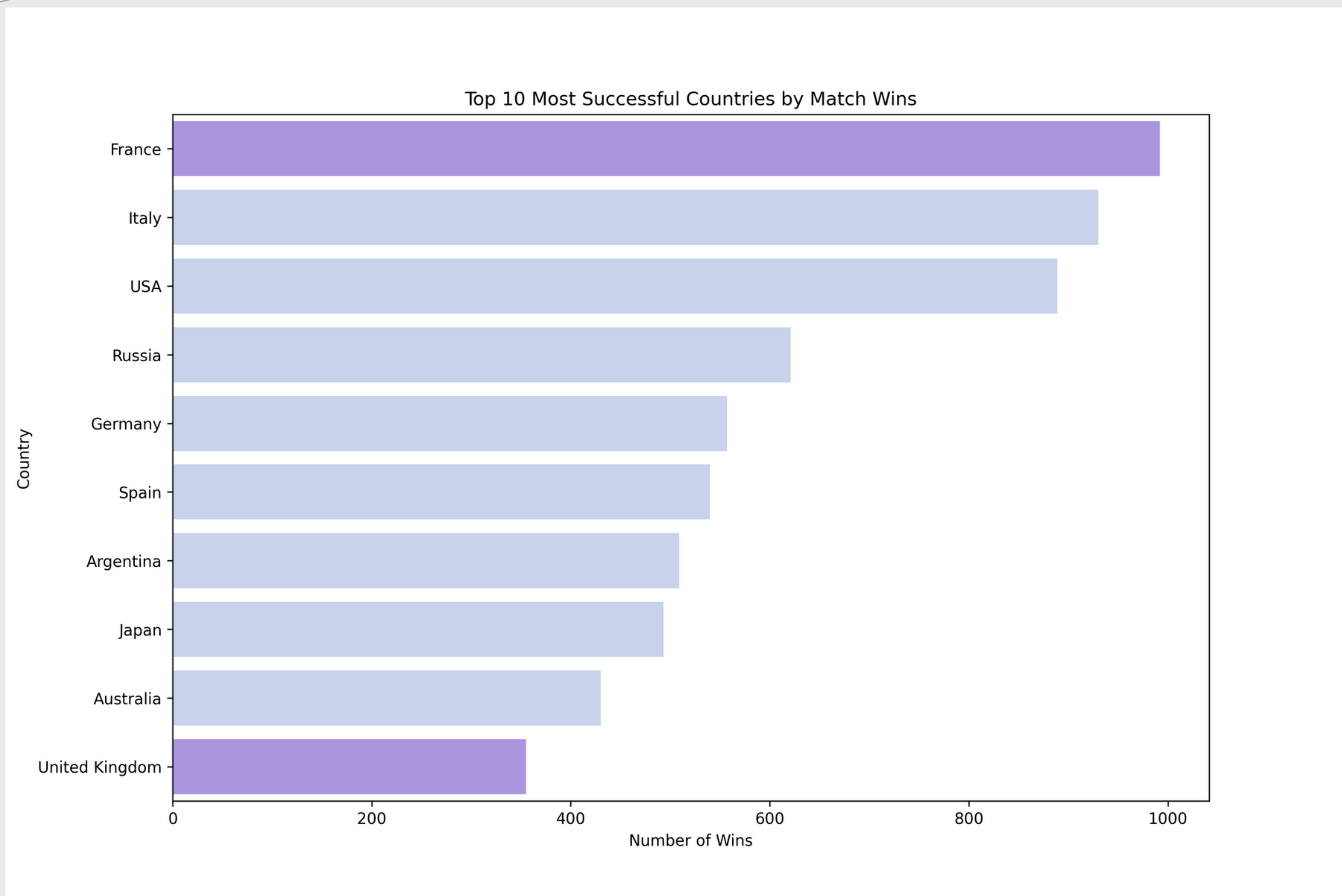
Spain **540**

Argentina **509**

Japan **493**

Australia **430**

United Kingdom **355**





Top 10 countries with the best (lowest) current ranks:

Poland **1.0**

Serbia **1.0**

Belarus **2.0**

Italy **2.0**

Spain **2.0**

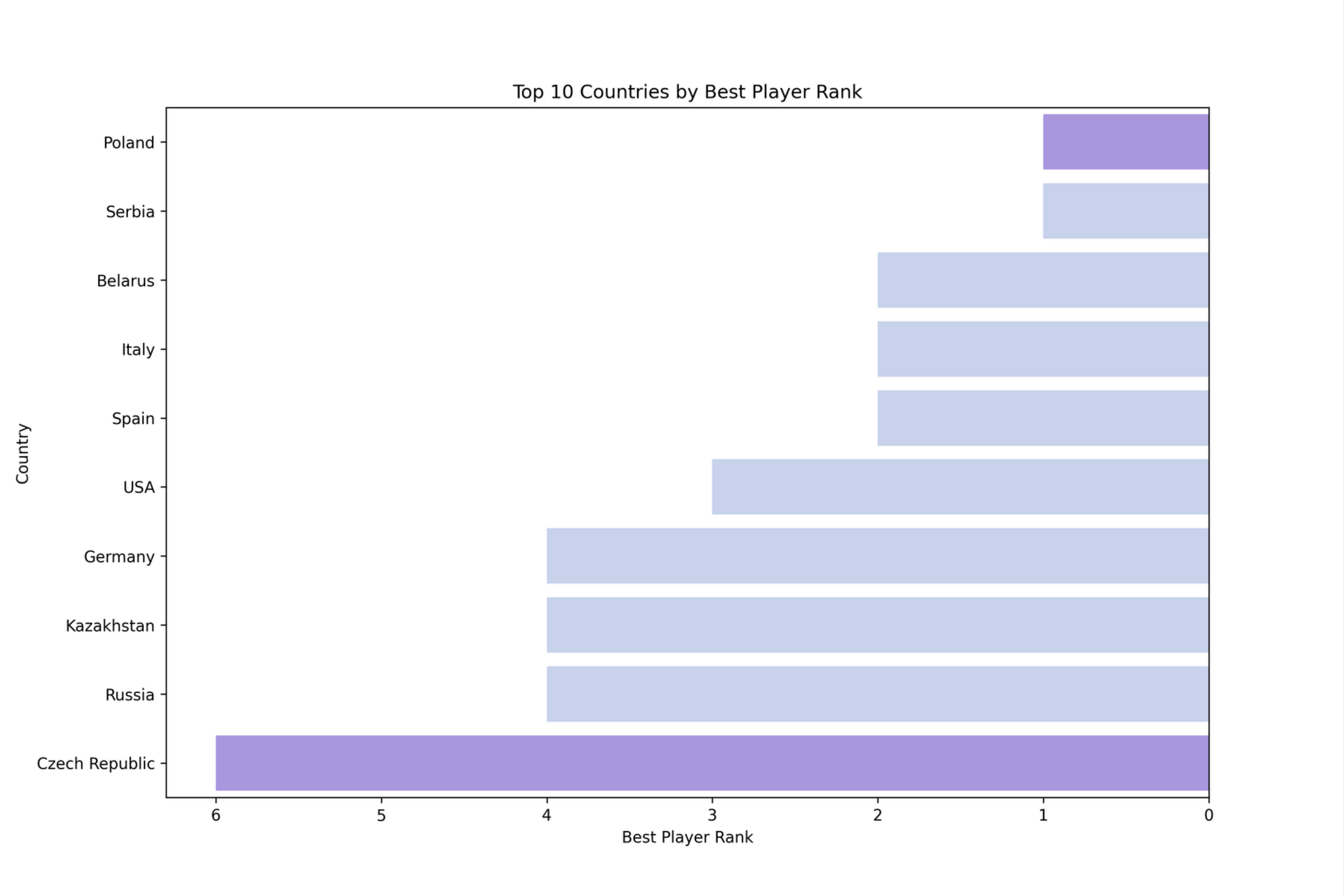
USA **3.0**

Germany **4.0**

Kazakhstan **4.0**

Russia **4.0**

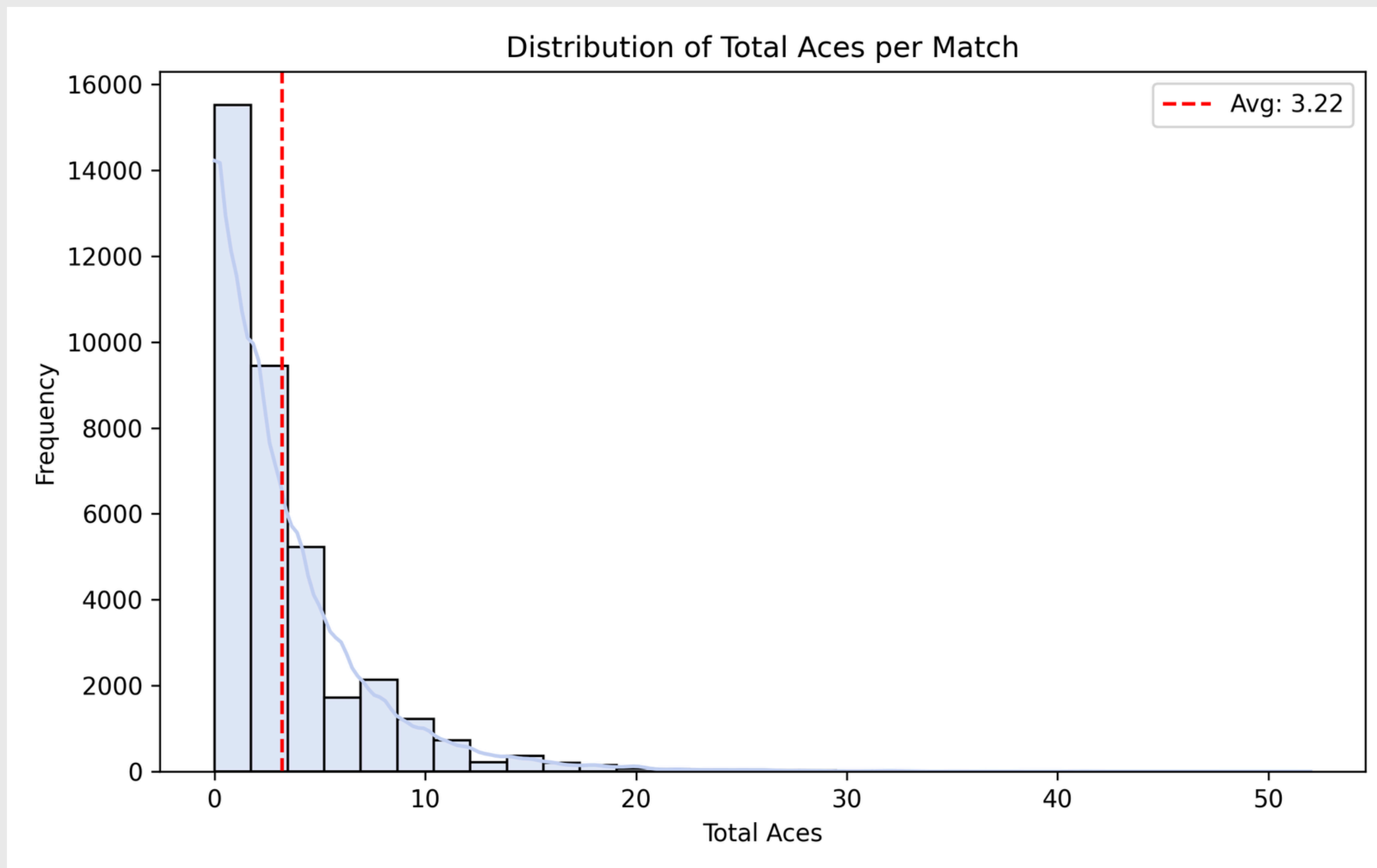
Czech Republic **6.0**

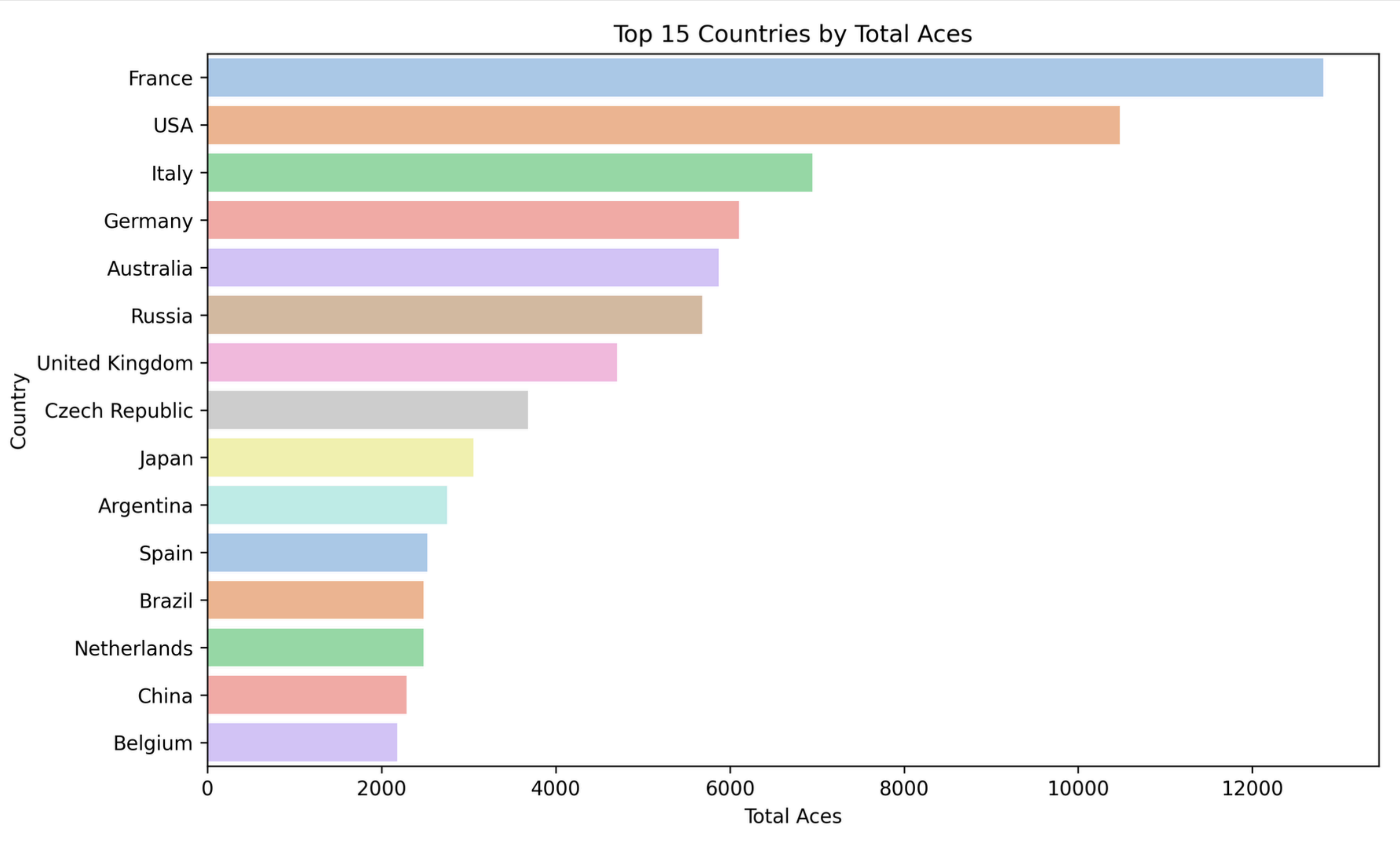


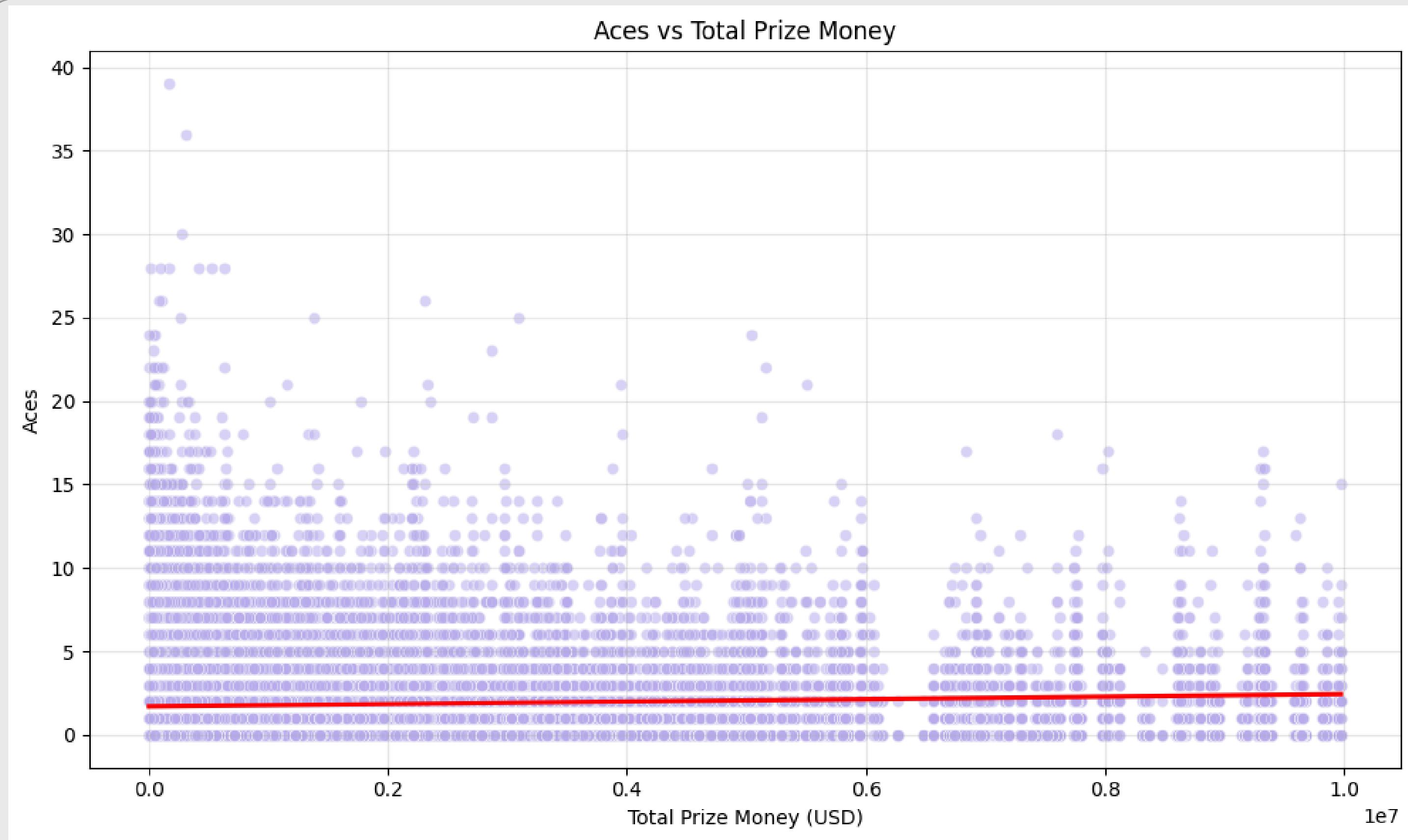
QUESTION 7 :

What is the average number of aces per match?

Average number of aces per match: **3.43**







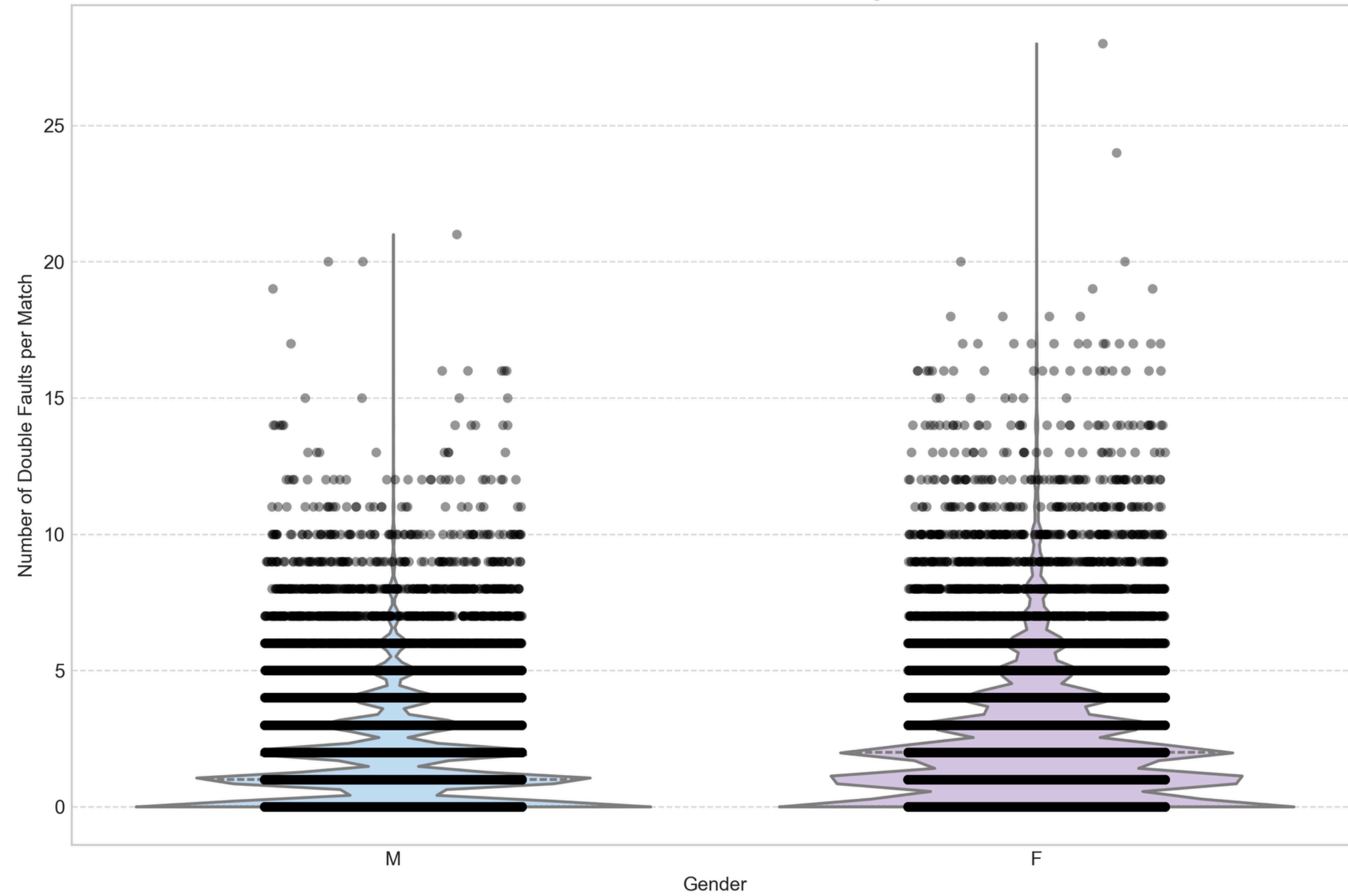
QUESTION 8 :

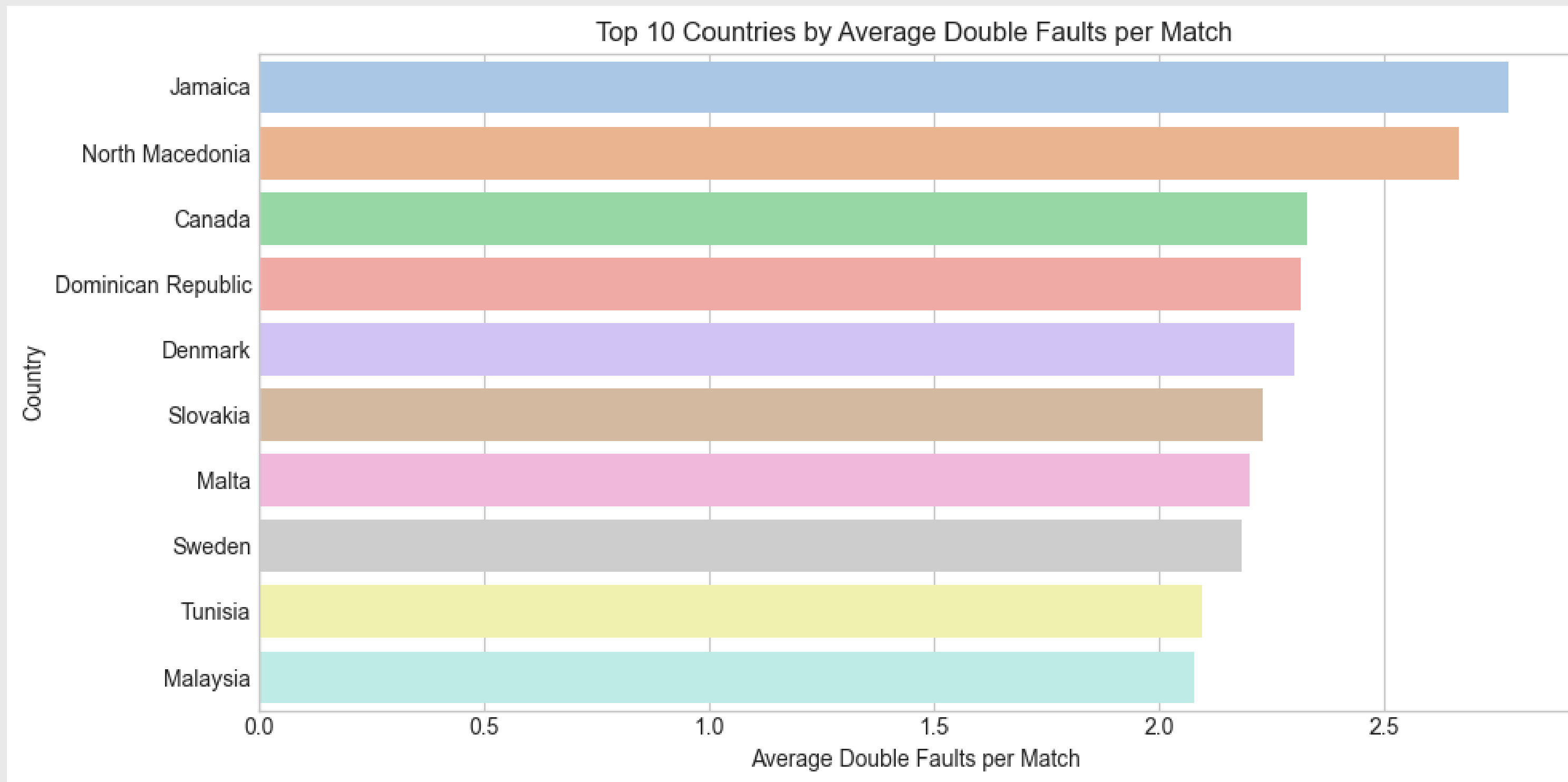
Is there a difference in the number of double faults based on gender?

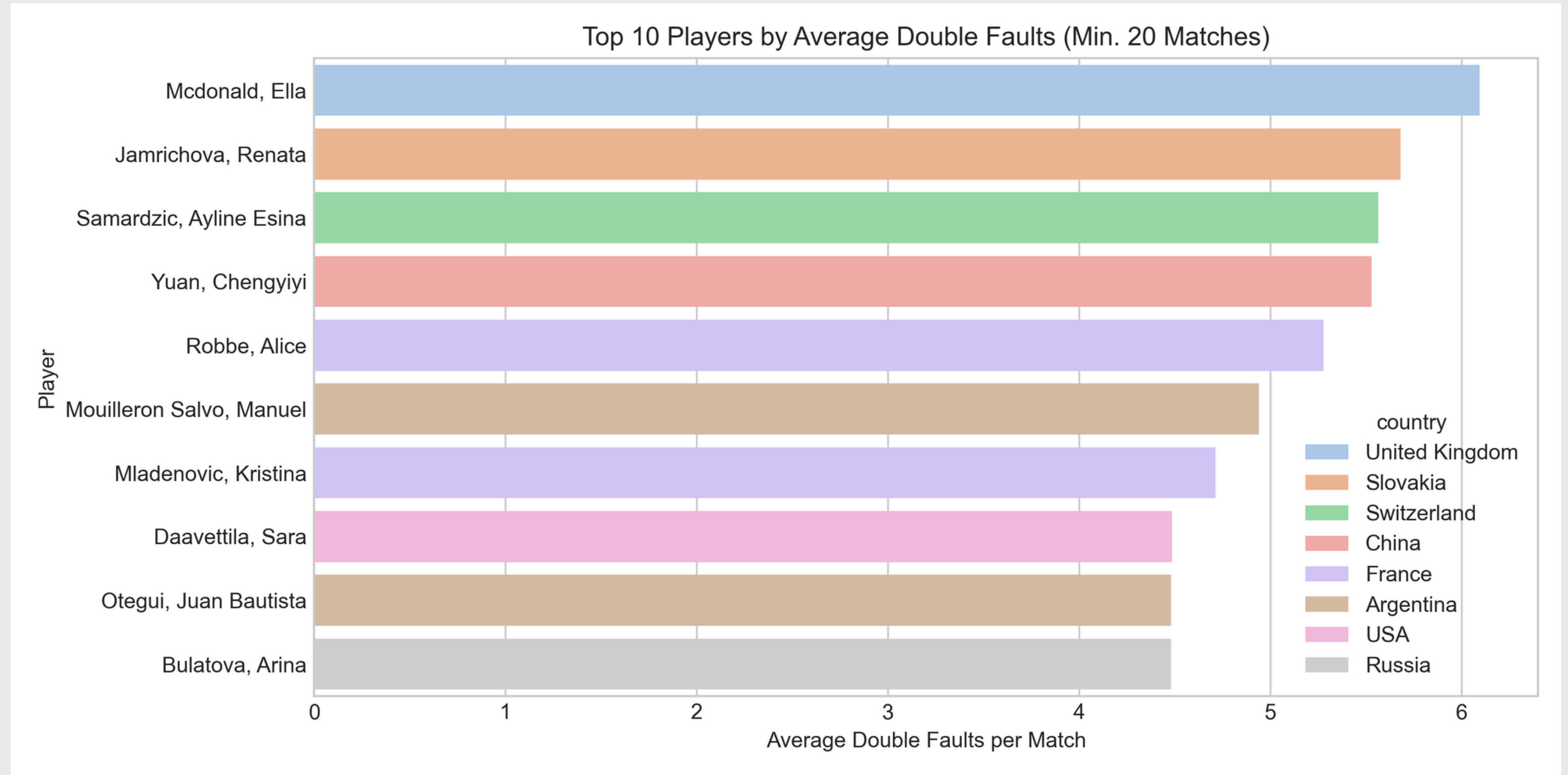
Descriptive Statistics by Gender:

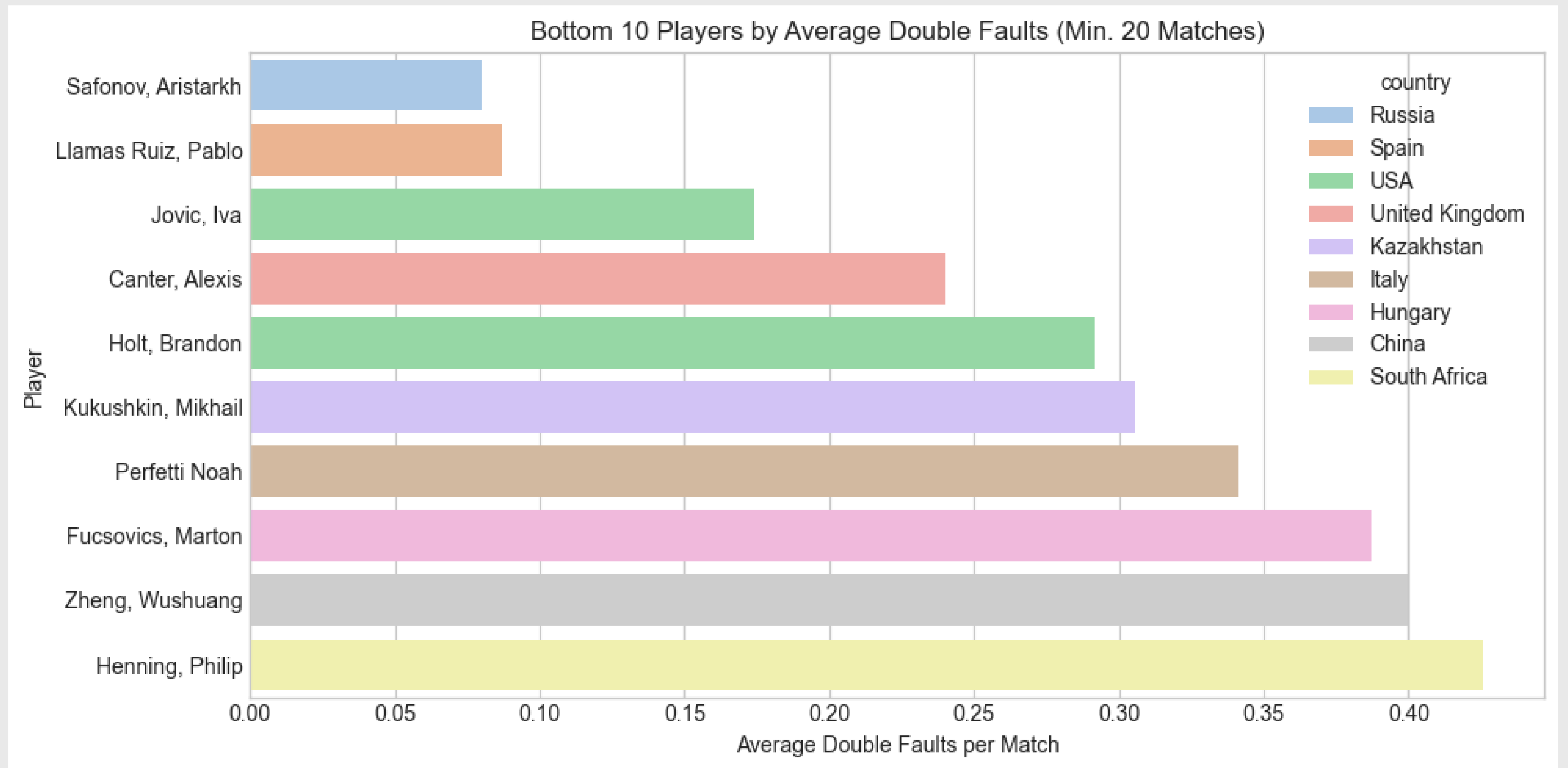
	count	mean	std	min	25%	50%	75%	max
F	30950.0	2.107981	2.249022	0.0	0.0	2.0	3.0	28.0
M	40166.0	1.599064	1.801070	0.0	0.0	1.0	2.0	21.0

Distribution of Double Faults by Gender









QUESTION 9 :

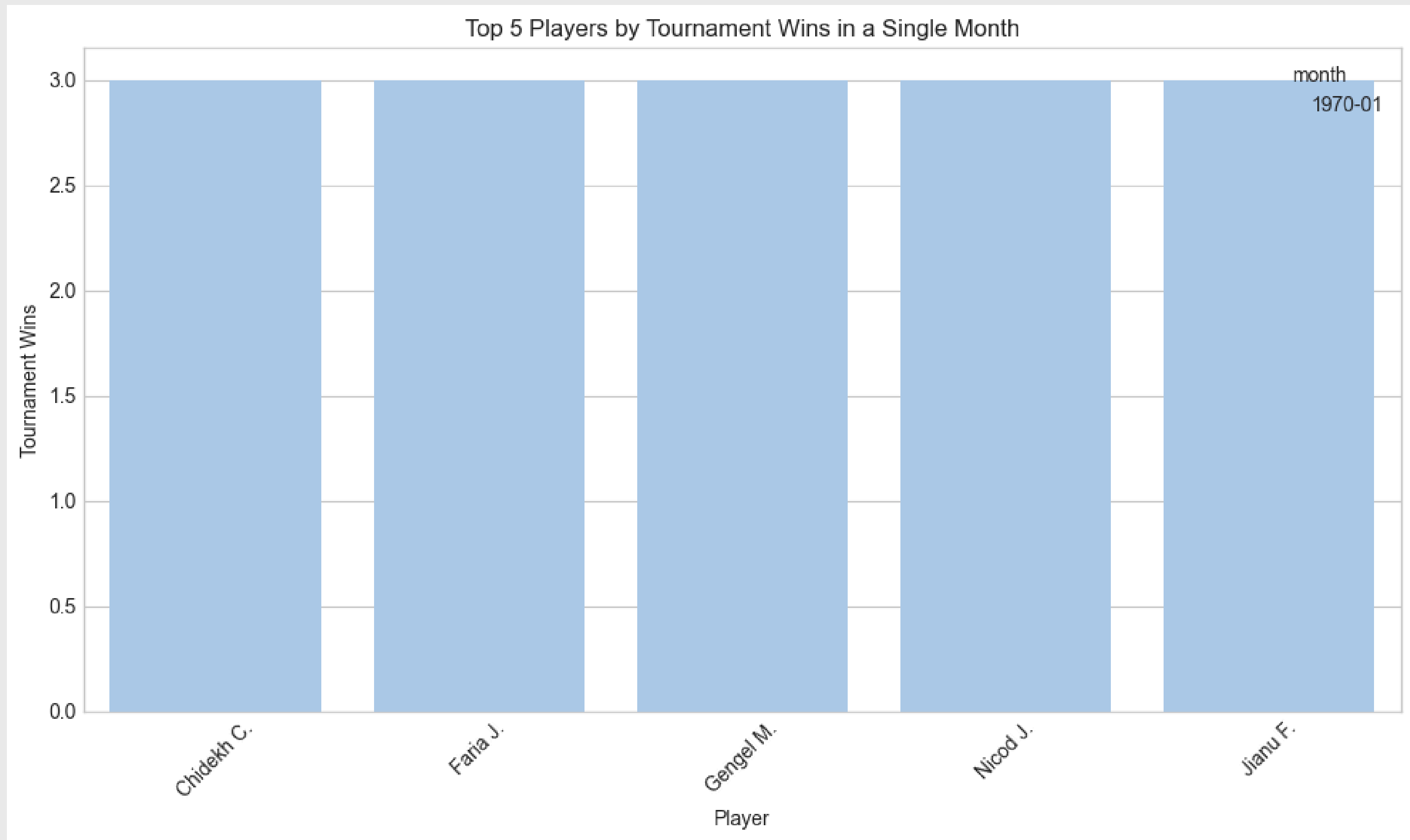
Which player has won the most tournaments in a single month?

Player with most tournament wins in a single month: winner

Chidekh C.

month 1970-01

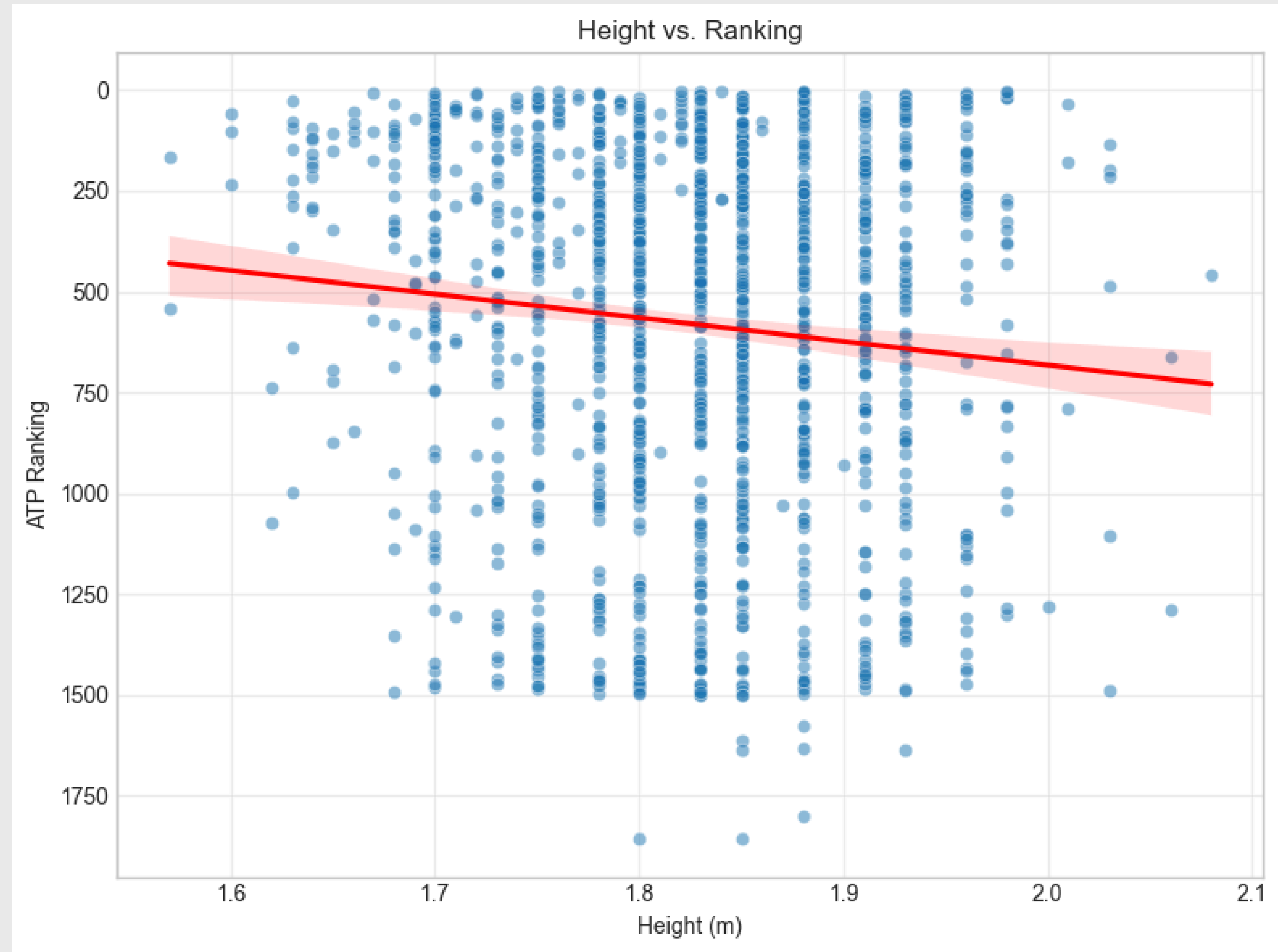
tournament_wins 3



QUESTION 10 :

Is there a correlation between a player's height and their ranking?

No, there's not a correlation between a player's height and their ranking.

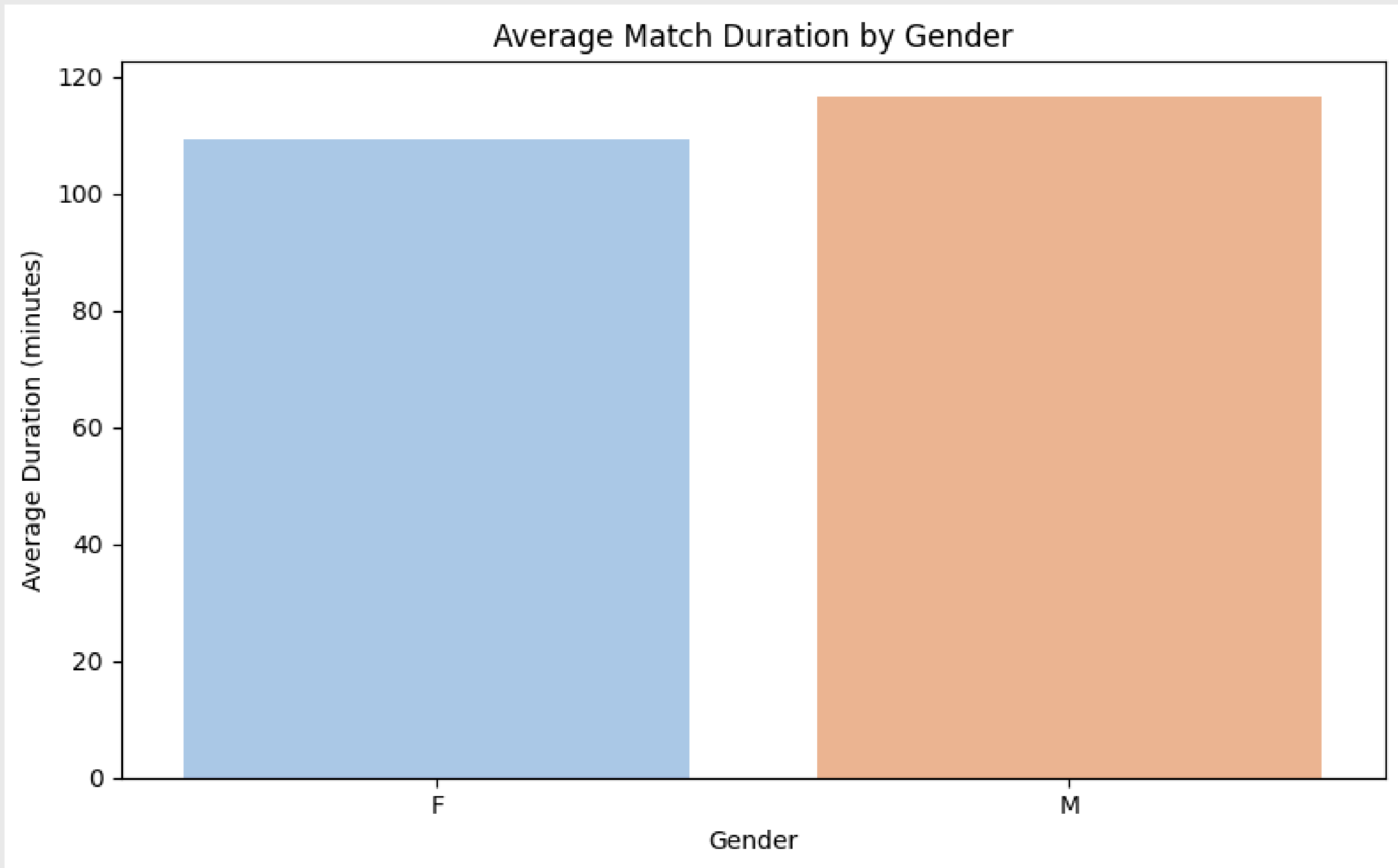


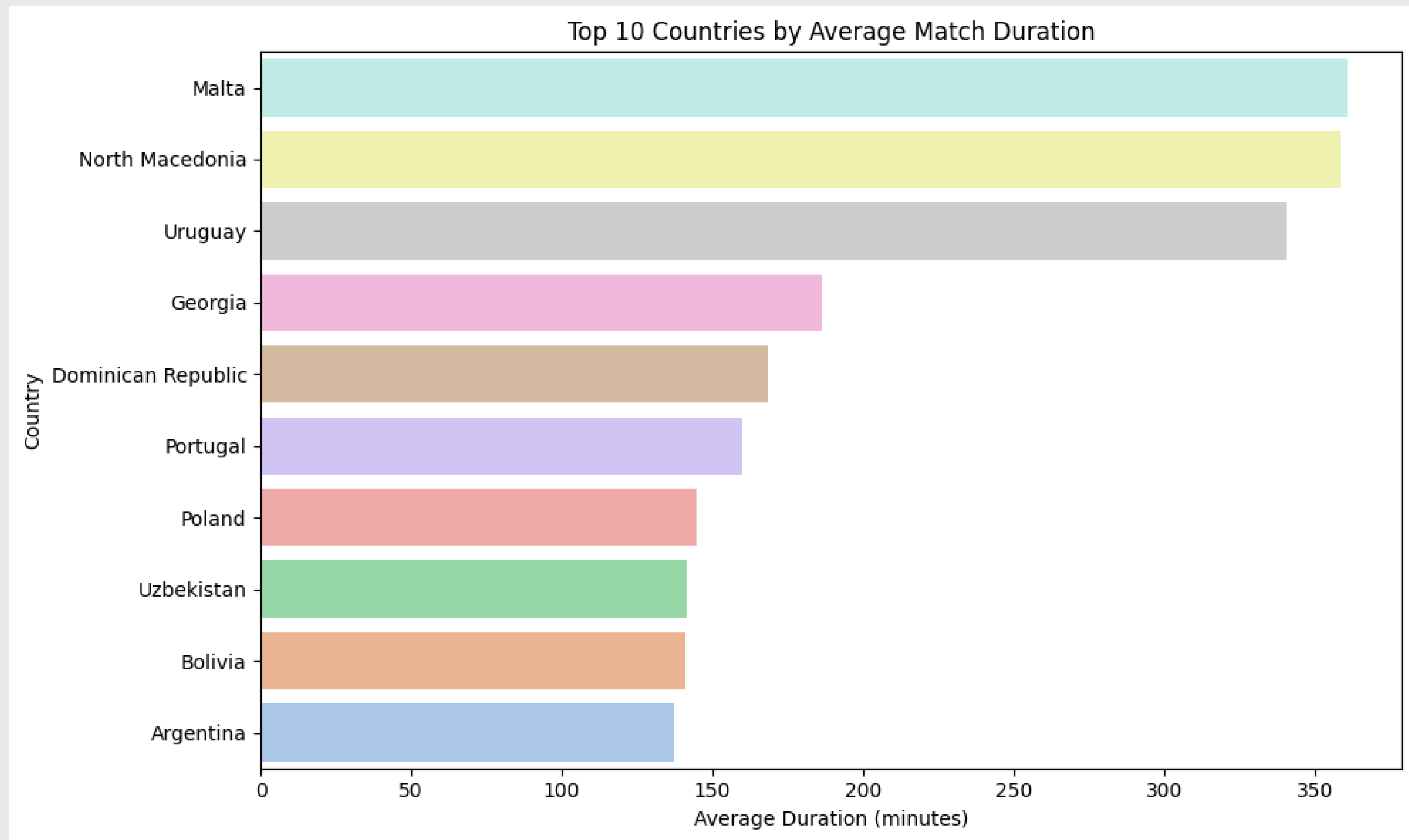
QUESTION 11 :

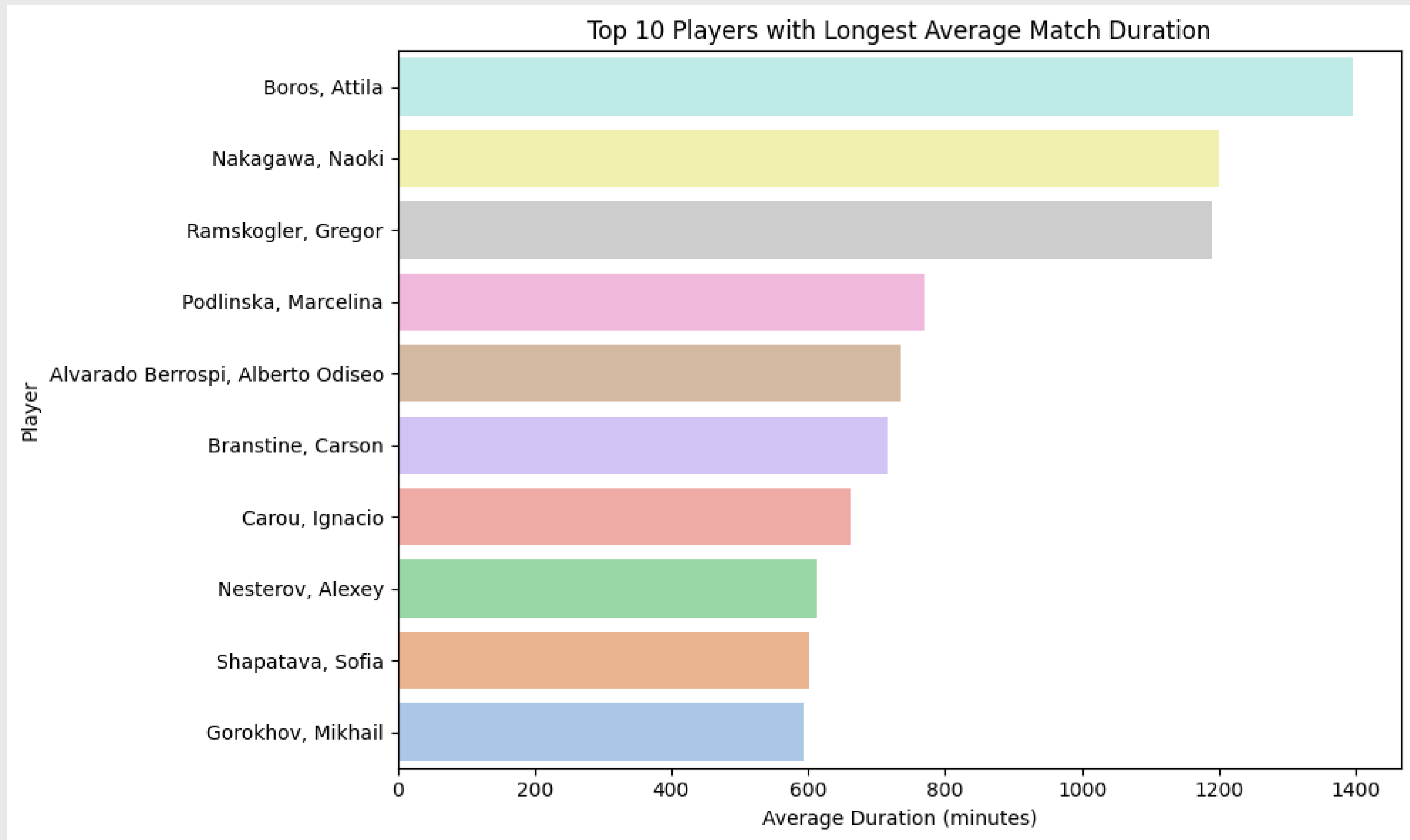
What is the average duration of matches?

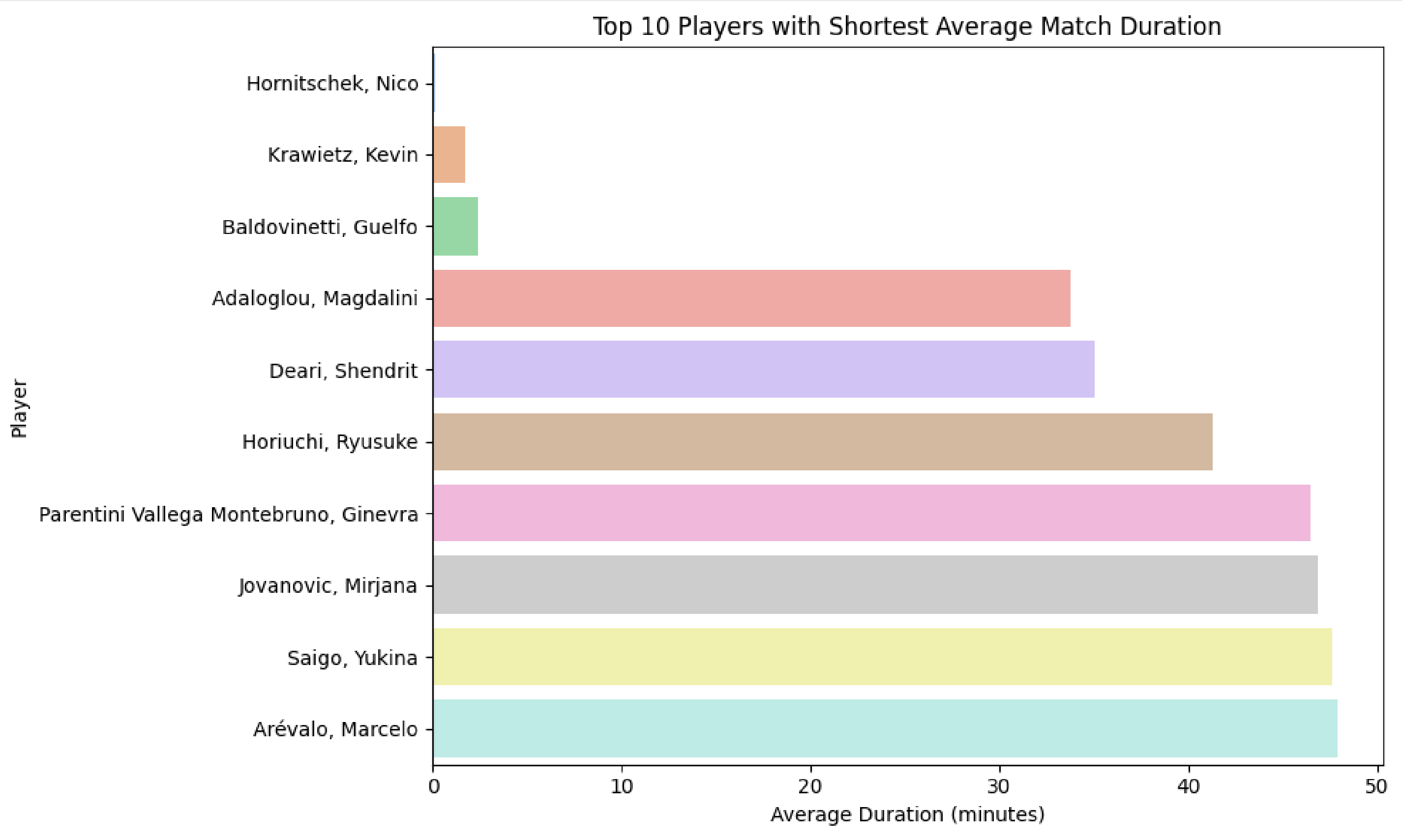
Average match duration (in minutes) by gender:

Gender	Total_duration_min
Female:	109.349740
Male:	116.743139







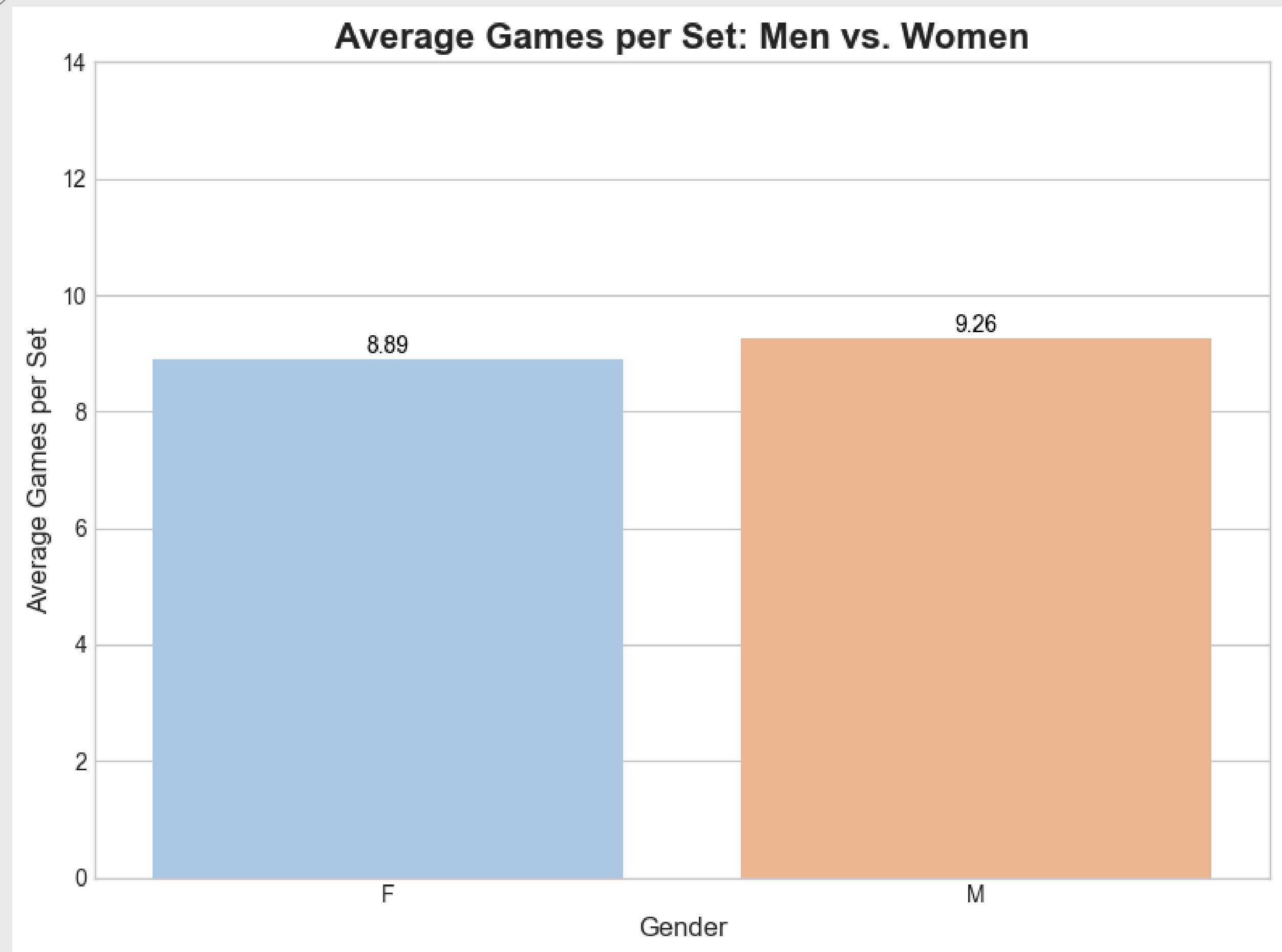


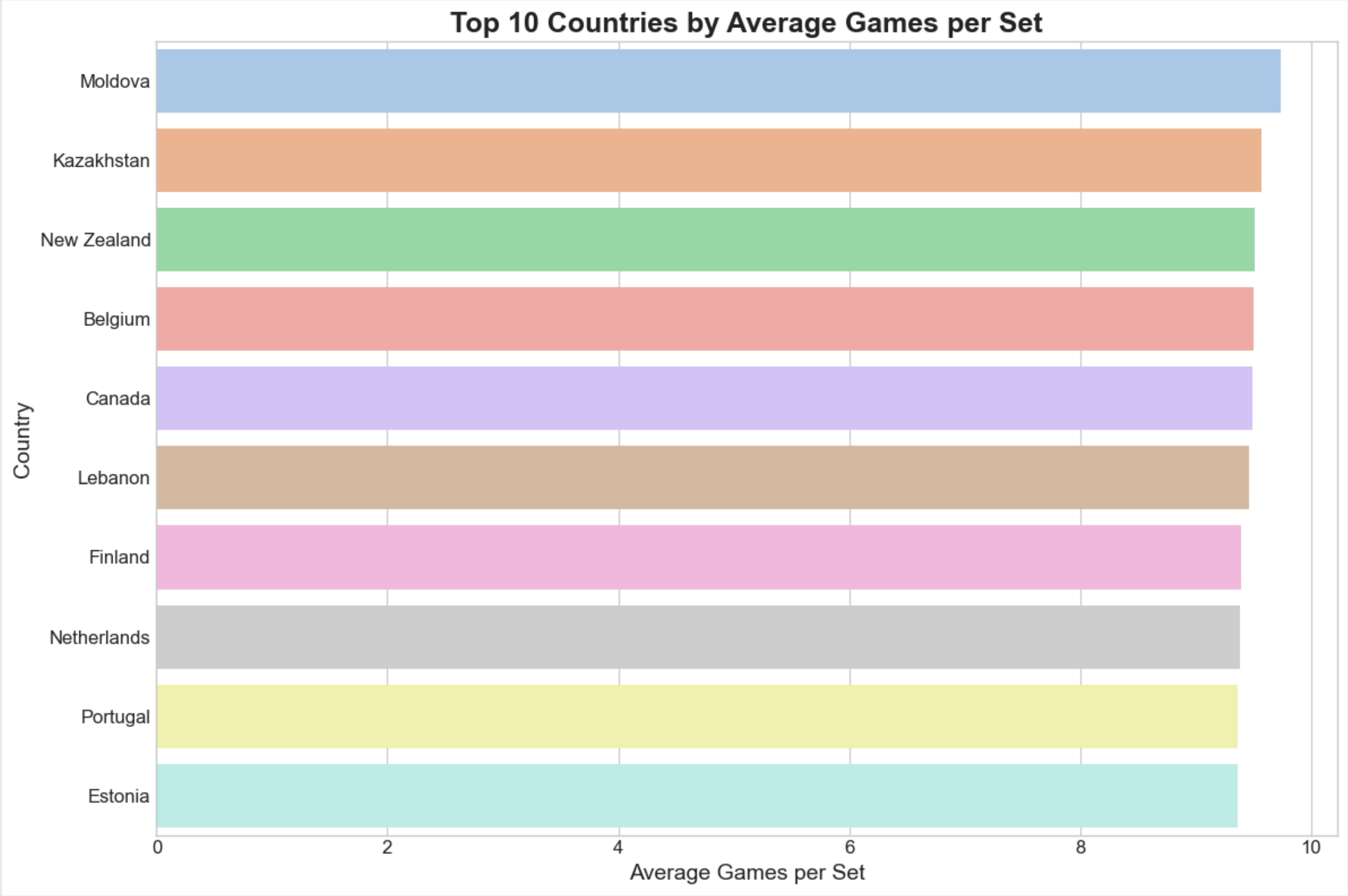
QUESTION 12 :

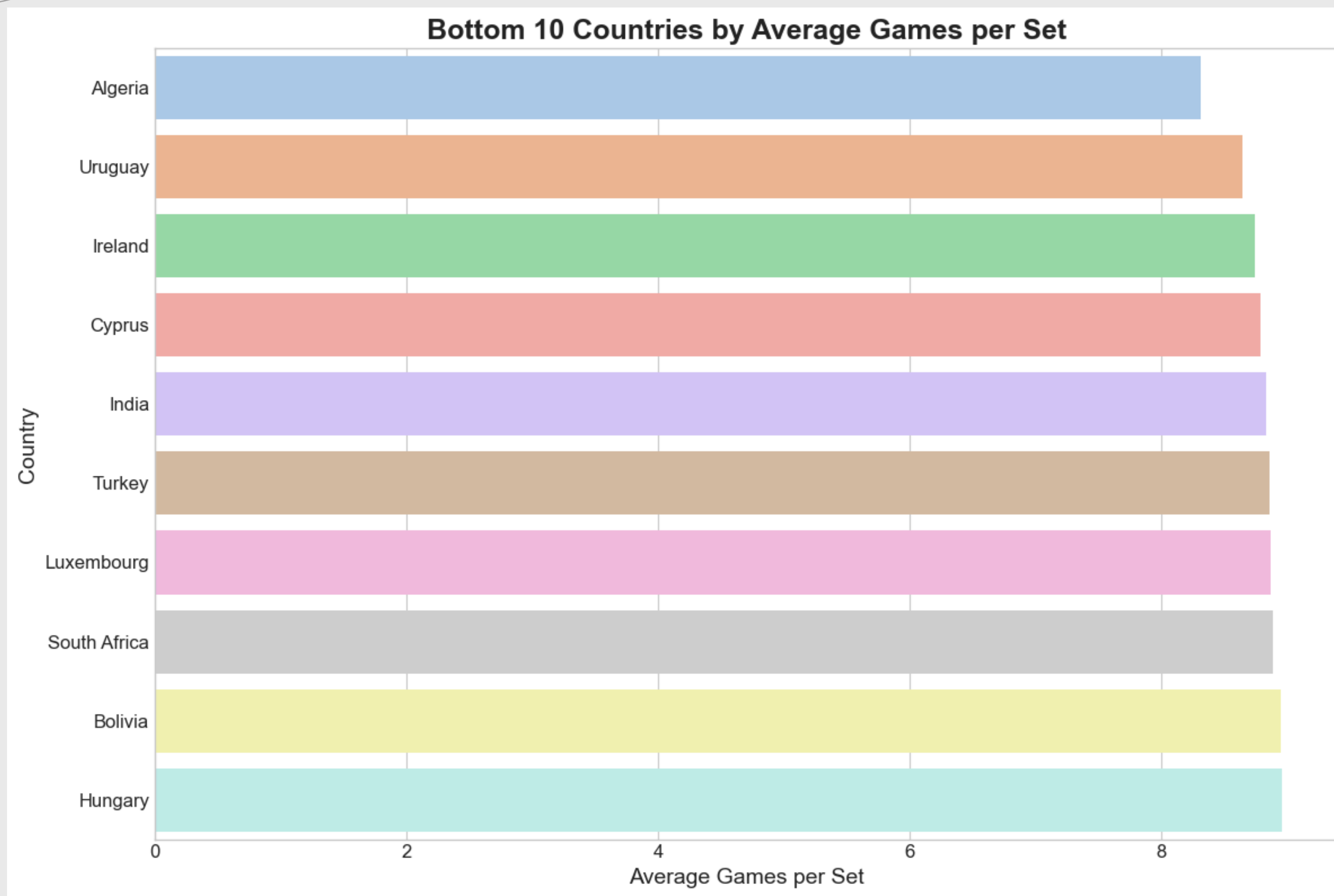
What is the average number of games per set in men's matches compared to women's matches?

In men's matches : 9.26

In women's matches: 8.89



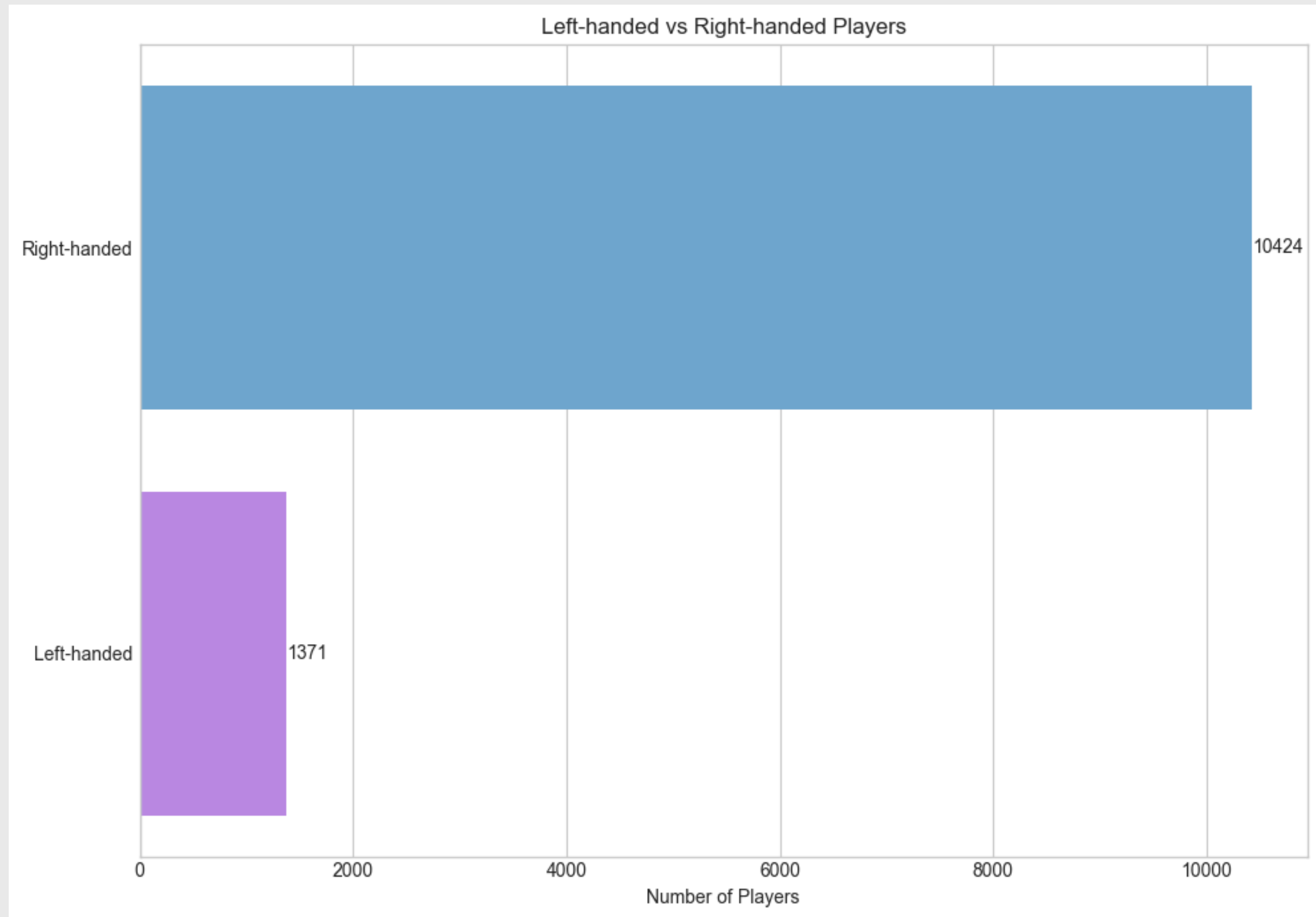




QUESTION 13 :

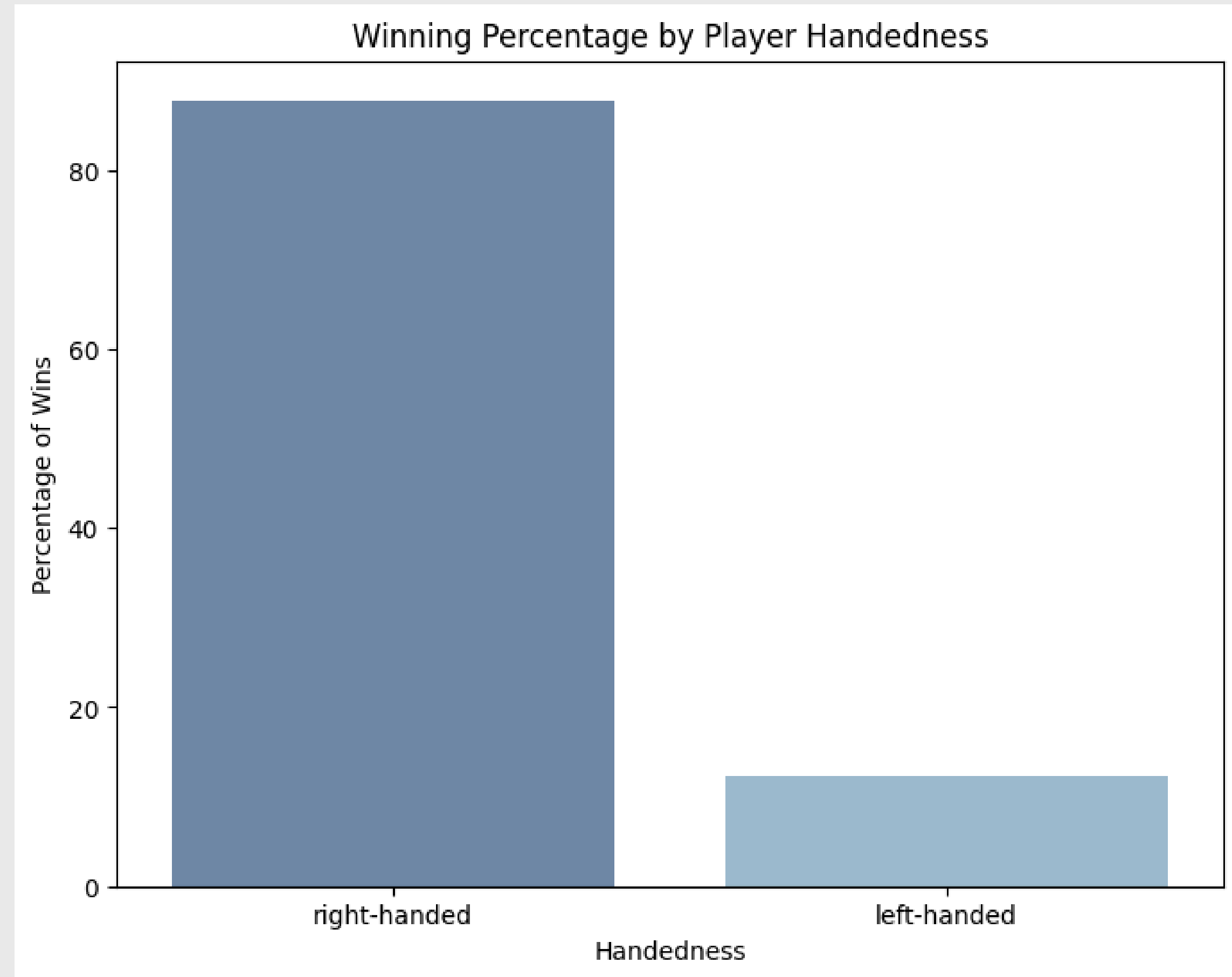
What is the distribution of left-handed versus right-handed players?

Right-handed Players: **10424**
Left-handed Players: **1371**





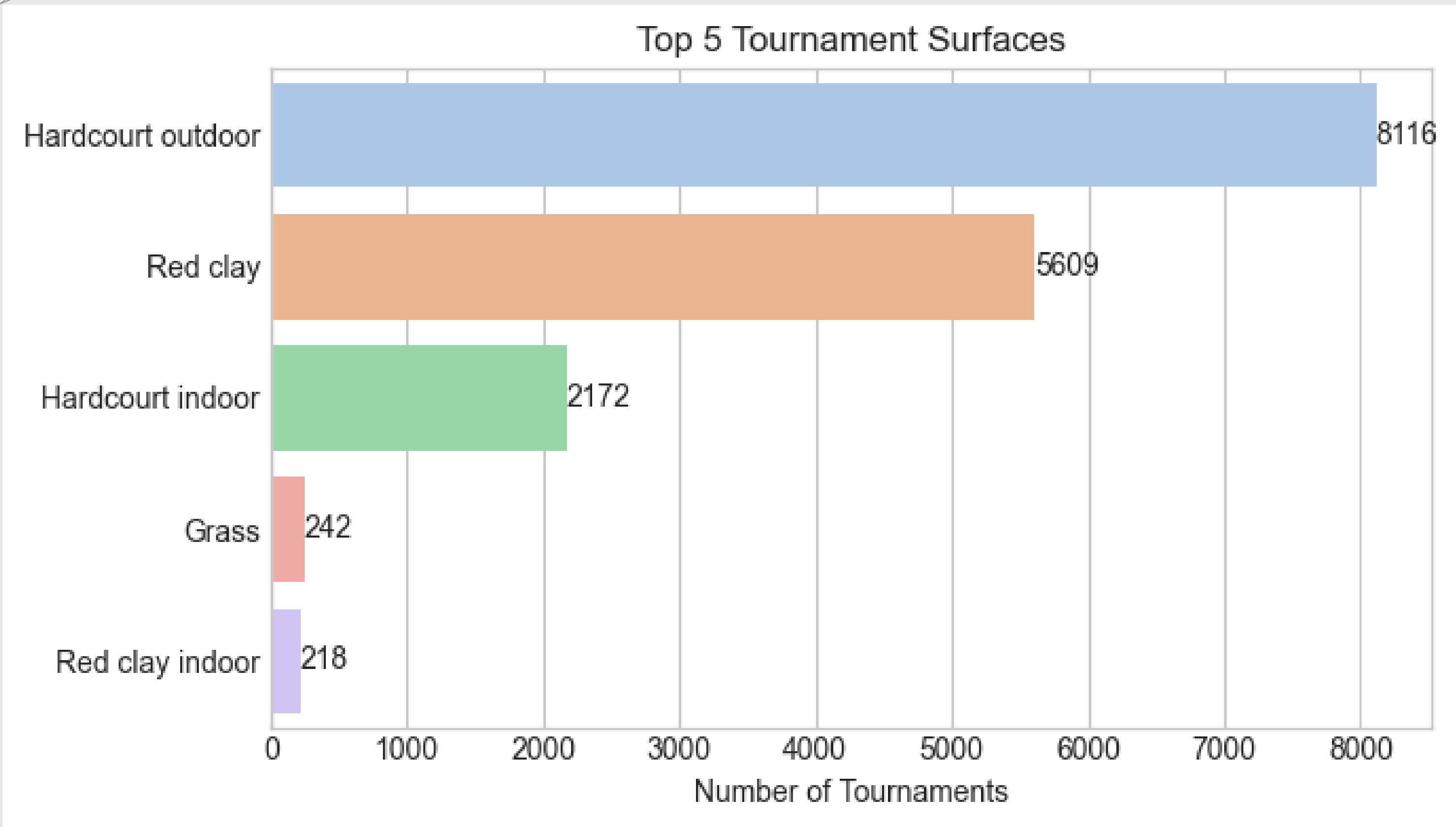
Players win based on:
Right-handed: **87.744843%**
Left-handed: **12.255157%**



QUESTION 14 :

What is the most common type of surface used in tournaments?

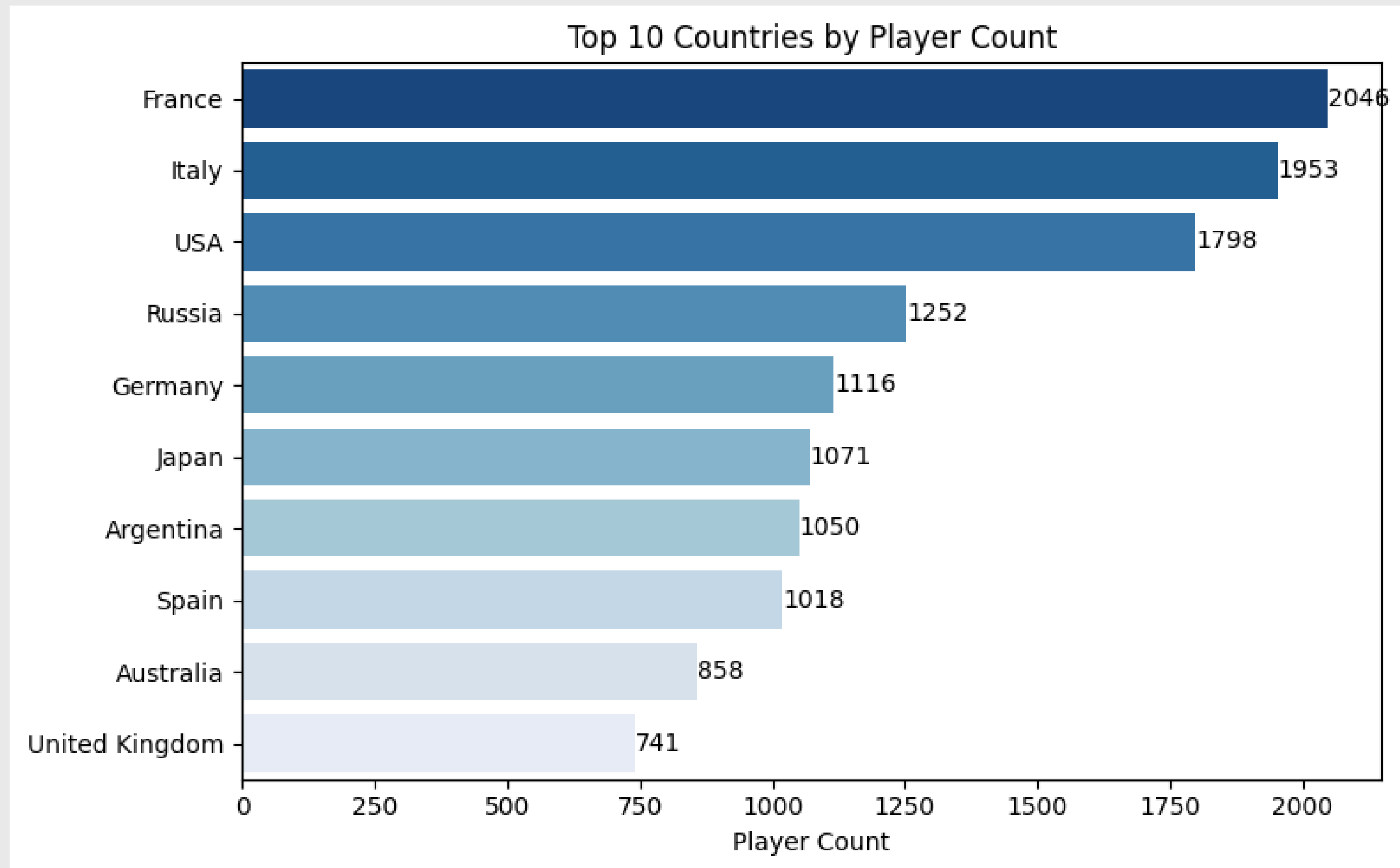
Top Surfaces :
Hardcourt outdoor
Red clay
Hardcourt indoor
grass
Red clay indoor



QUESTION 15 :

How many distinct countries are represented in the dataset?

The number of distinct countries is: **101**

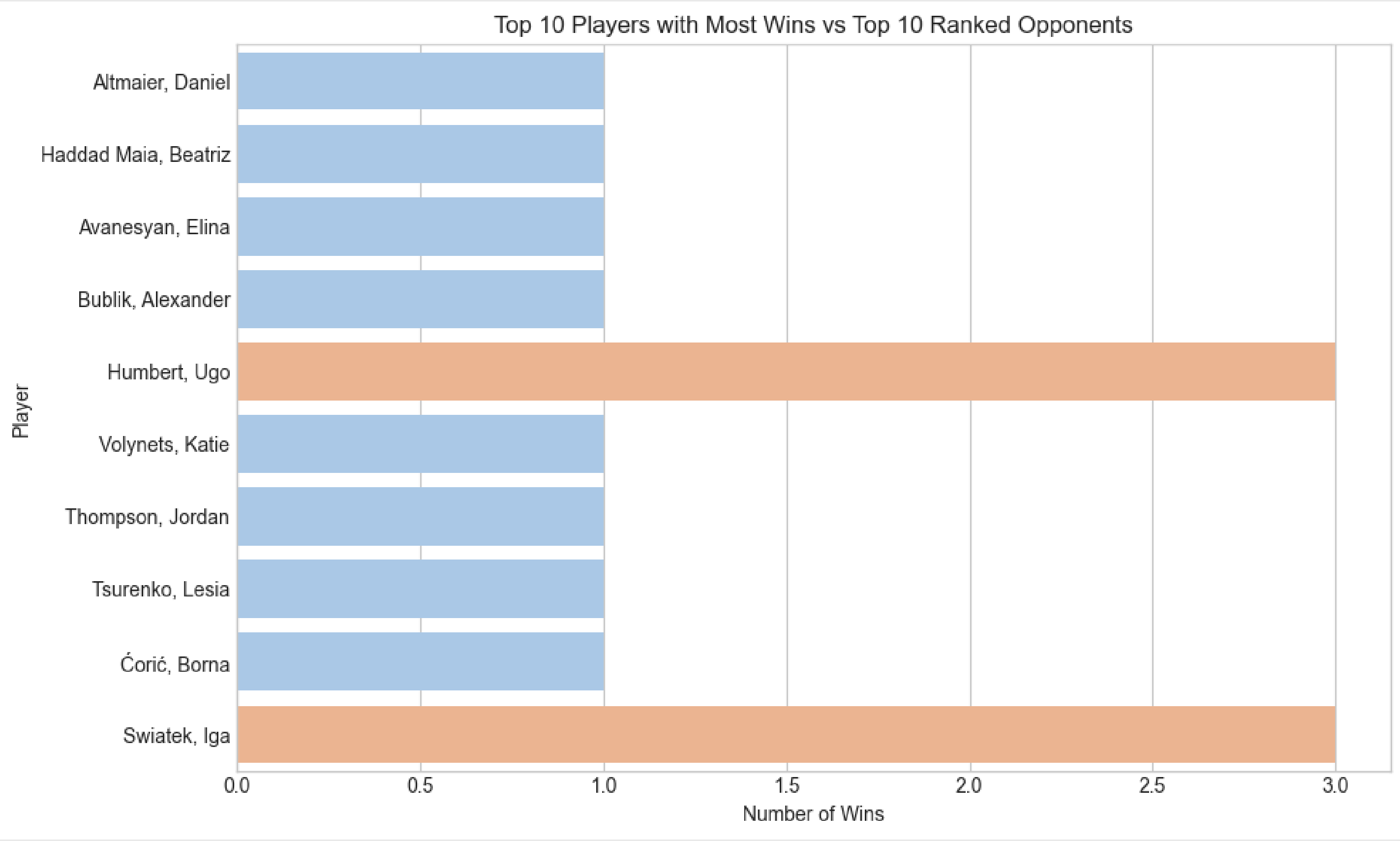


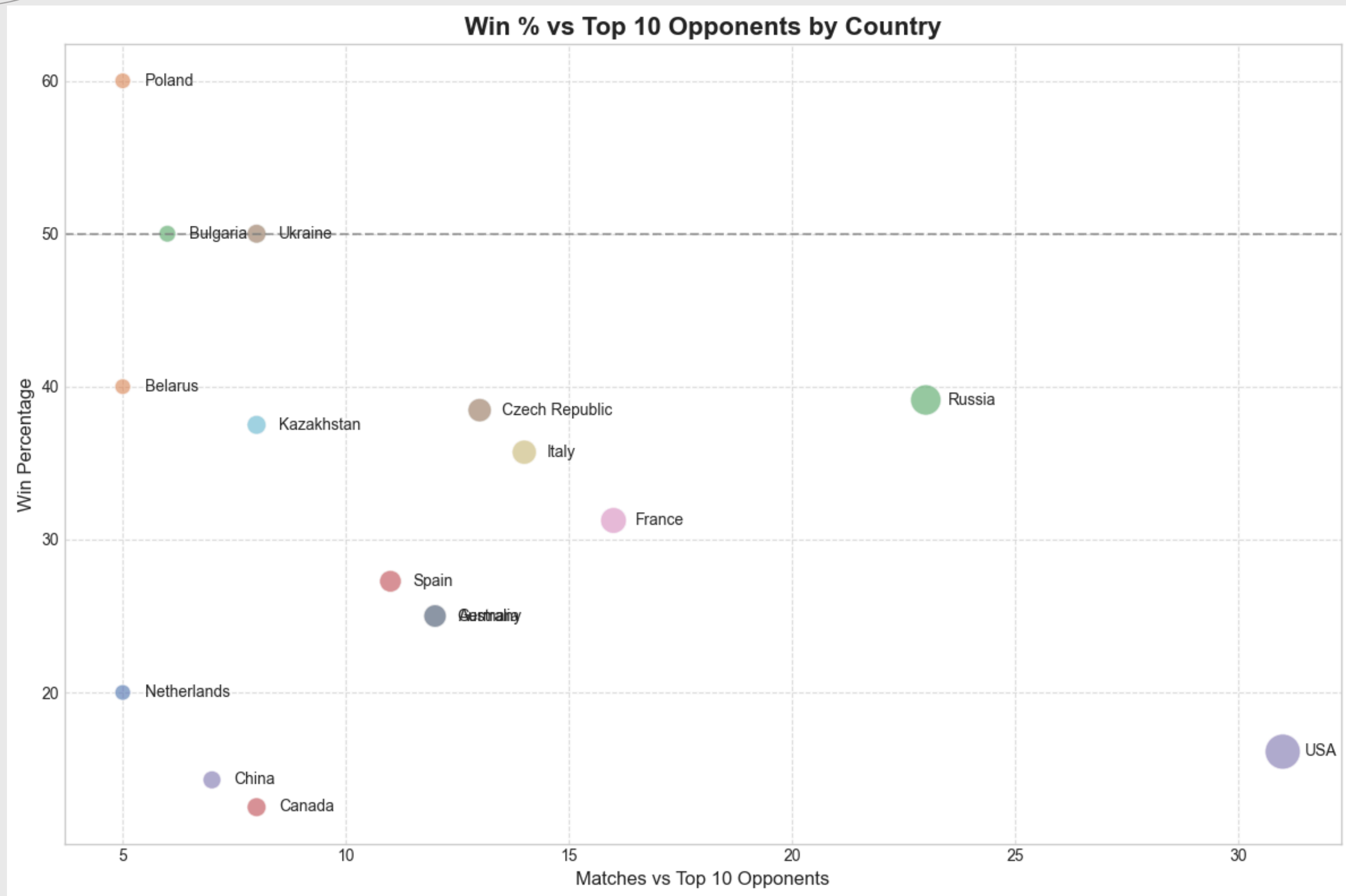
QUESTION 16 :

Which player has the highest winning percentage against top 10 ranked opponents?

Right-handed Players: **10424**

Left-handed Players: **1371**

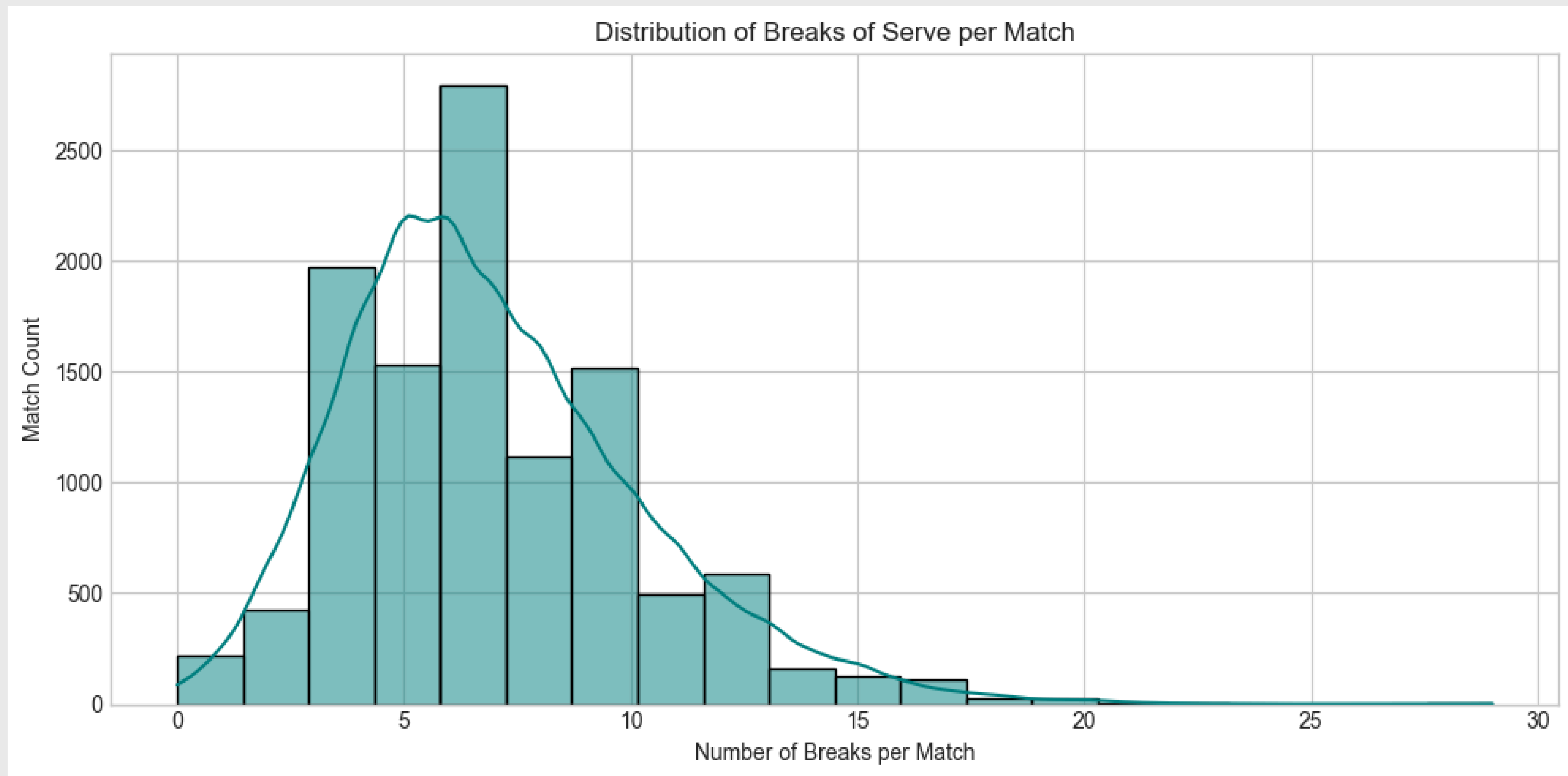


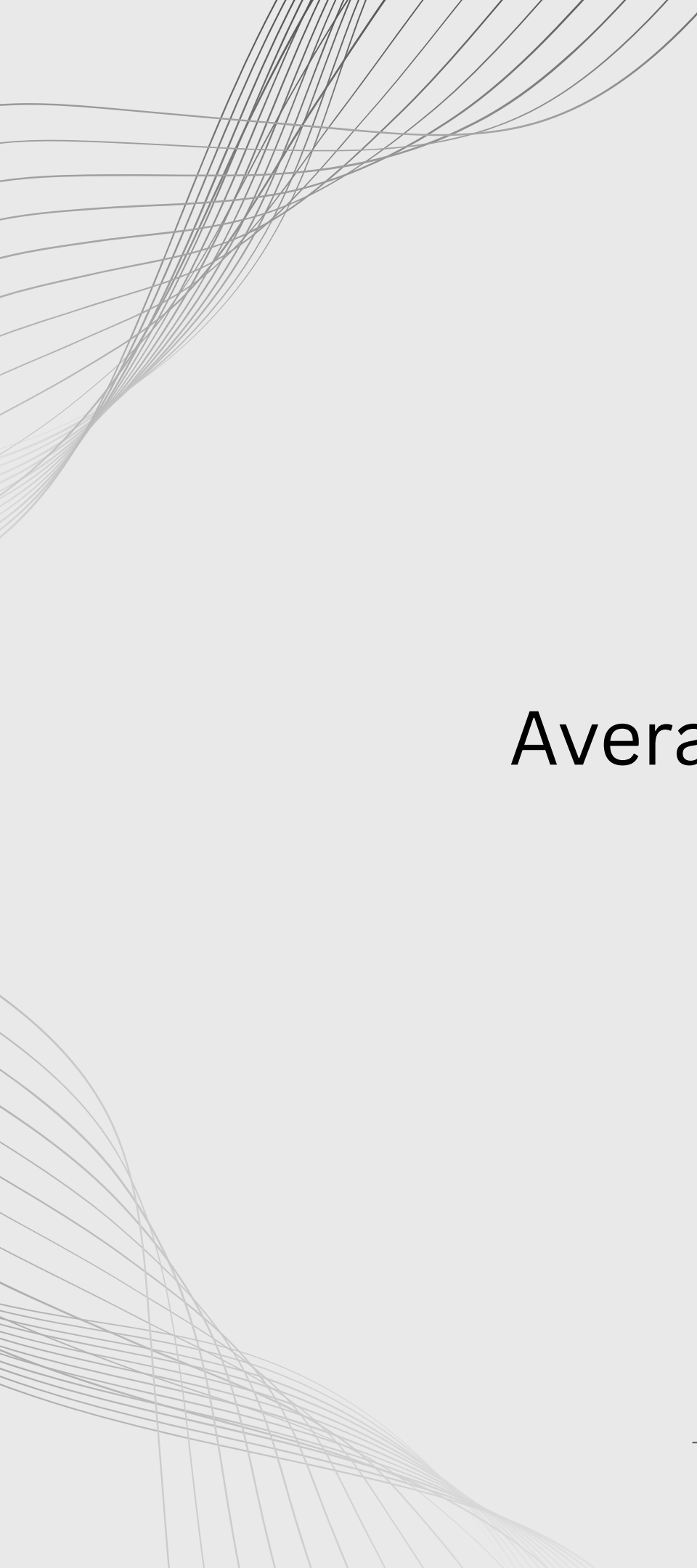


QUESTION 17 :

What is the average number of breaks of serve per match?

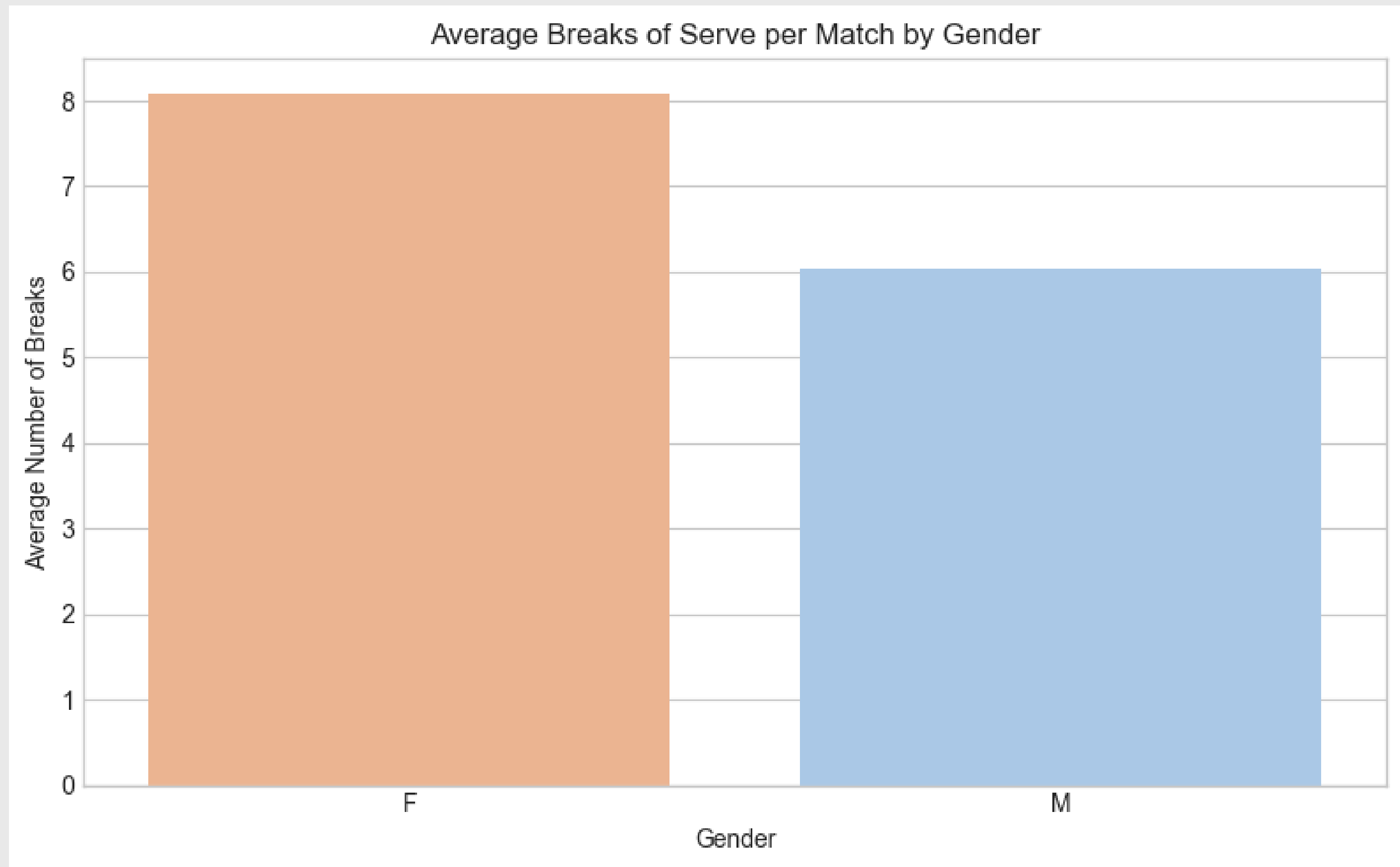
Average Number of Breaks per Match: **6.93**





Average Number of Breaks per Match by Gender:

Gender	num_breaks
Female	8.083681
Male	6.033898





THANK YOU FOR YOUR TIME

LINK TO THE CODE:

[HTTPS://GITHUB.COM/MARZI-DEZFOULIANRAD/DATA_ANALYSIS_TENNIS](https://github.com/marzi-dezfoulianrad/data_analysis_tennis)
